

CUBE 30 TOUCH



SERVICE MANUAL

Rev. 1.3 – July 2021

Automatic instrument for ESR determination
with modified Westergren method

Software version 1.xx.xx

FOR IN VITRO DIAGNOSTIC USE ONLY

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CUBE 30 touch models:

Catalog number	Description
10395	CUBE 30 touch

Manual revisions list:

MANUAL revisions	Description of changes
1.0 of 02/2018	Official Version
1.1 of 07/2018	Alignment with SW V1.13
1.2 of 08/2018	Spare parts review
1.3 of 07/2021	Update due to: - new graphic design; - new components in 3d printing; - new plastic protections; - new colours for samples holders.



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Symbols:

Key of graphic symbols	
	Instrument meeting the requirements of the European Directive 98/79/EC on in vitro diagnostic medical devices.
	In vitro diagnostic medical device
	Manufacturing date
	Serial number
	Manufacturer
Key of electric and safety symbols	
	Protective conductor
	WEEE: Waste electrical and electronic equipment – Separate Collection is required pursuant to Legislative Decree No. 49 of March 14. 2014 (Italy), on “implementation of Directive 2012/19/EC.”
	Attention: please read this manual carefully and comply with the safety symbols.
	Caution: risk of electrical shock
Key of graphic symbols used in this document	
	WARNING: potential risk of personal injury; all the conditions indicated in the relative text must be read and understood before proceeding.
	CAUTION: potential risk of damage to the instrument; all the conditions indicated in the relative text must be read and understood before proceeding.
	N.B. important information.
	BIOHAZARD: risk of contamination with potentially infected substances.
	Instrument that complies with MET standards for the Canadian and US markets



LIMITATIONS AND WARNINGS

Revision 1.3 of this manual corresponds to the CUBE 30 touch model of the instrument with the 1.xx family of software installed. It was drafted and carefully reviewed and this version is closely related to the instrument model (data can be obtained from the instrument's ID plate) and the version of the software that controls it (data can be obtained through a procedure on the instrument itself).

It must be read carefully before using the instrument, especially the parts relating to safety.



ATTENTION

Do not use this manual except in its complete form.

If this manual is used in an incomplete form, DIESSE Diagnostica Senese S.p.A. declines any responsibility for adverse results.



ATTENTION

Any instrument updating done with the customer's authorization requires that the user manual be updated in a corresponding manner.

DIESSE Diagnostica Senese SpA accepts no responsibility for damage caused directly or indirectly by errors, defects, or incidents due to any use of this manual that does not correspond with the version of the supplied instrument.

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1 GENERAL DESCRIPTION OF THE INSTRUMENT

1.1 Overall view of the unit

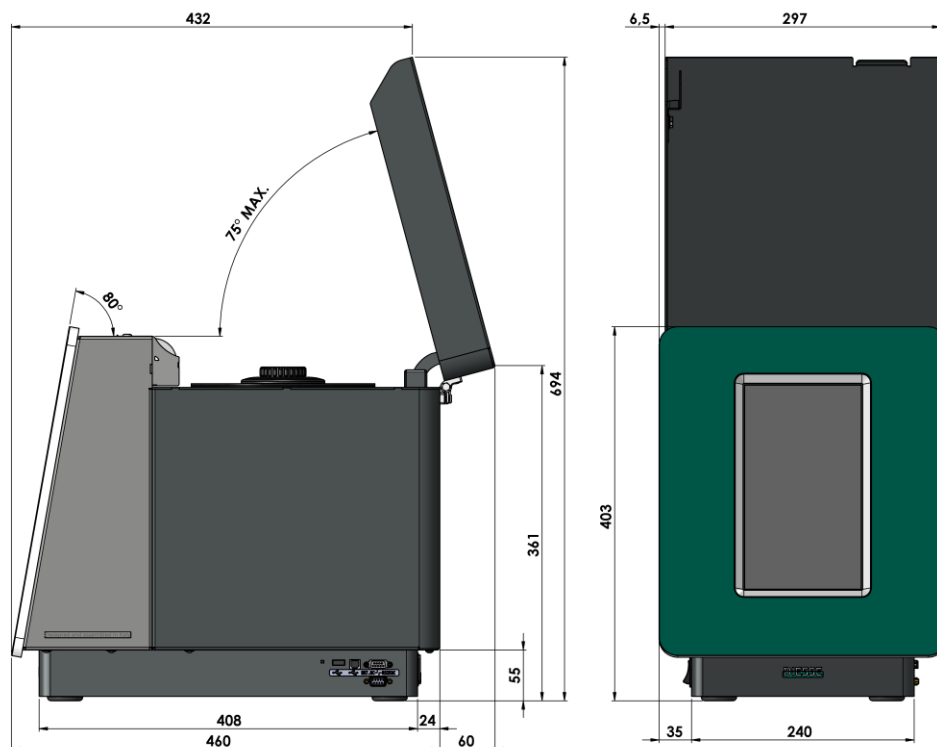


Figure 1.1

The CUBE 30 TOUCH module consists of a single body containing all the operating functions necessary for analyzing the sample.

The instrument has 30 housings arranged in two concentric circles with 15 positions each, which allow the insertion of samples. The tube rack is also equipped with a reducer for the use of rubber tubes. (Use the plate reducer only when using rubber tubes)
One motor lifts the reading unit which utilizes four optical sensors (one per position) to verify the suitability of the sample and detect the level at baseline and at the end of the sedimentation time.

Sample identification is carried out using an internal barcode reader that reads when the sample is detected by the optical sensor. The sample identification code can also be entered manually using a virtual keyboard if not automatically read by the instrument.

The function of the alarm is to alert the operator during various stages of the operating cycle or in the event of errors.

Internal Temperature Sensor: Used to measure the temperature and is placed near the tube tray

The printer Prints test results at the end of each work cycle.

External connections

On the rear side of the instrument, there is a power supply connector.



Figure 1.2

On the left side of the instrument, there is the ON/OFF power button.



Figure 1.3

On the right side of the instrument, there are: two connectors for connection with Computer / HOST. One RS232 Serial type (9pin) and the other USB type. A connector for connection to an external barcode reader (EXT). BC). One USB port for USB mass storage connection (usable for software update or file export).

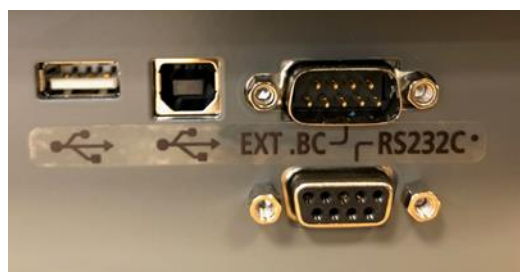


Figure 1.4

1.2 Technical specifications

USE	Internal use
POWER SUPPLY	Europe: 230Vac@50Hz; Usa/Canada: 110-120Vac@60Hz Power: 100VA
DIMENSIONS (mm)	310x470x403 (l x w x h)
WEIGHT	15 kg
ROOM TEMPERATURE	Operating temperature From +5° to +35°C Storage temperature From +5° to +45°C
RELATIVE HUMIDITY (RH)	20%-80% without condensation
ALTITUDE	Maximum 2000 meters (6562 ft)
NOISE LEVEL	below 80 decibel
LEVEL OF POLLUTION	2 pollution degrees
MEASUREMENT RANGE	from 0 to >140 mm/hr
CENTRAL UNIT	ARM Cortex-M4 180 MHz Microprocessor
DISPLAY	10,1" wide display with Resistive Touch Panel
OPTICAL UNIT	2 pairs of optical elements (photodiode + phototransistor)
INTERFACES	USB Host; USB Client; 2x RS232
PROTECTION CLASS	I
SAFETY STANDARDS	IEC 61010-1; IEC 6101-2-081; IEC 61010-2-101 CSA C22.2 No. 61010-1-12 / UL 61010-1:2012
EMC	EN 61326-1/EN 61326-2-6
INSTALLATION CATEGORY	II



Safety requirements and functional performances of the instrument are not assured if you use a different power supply from the one supplied in the package.

1.3 Installation

Please refer to the User Manual for the installation procedure.

1.4 Shutting down and shipping



Before shutting down and shipping the instrument it is recommended carrying out the following sanitizing procedure:

- A) The instrument must be turned off and cleaned internally of all residues or spillage with a liquid detergent and left to dry.
- B) Use one of the ready-to-use spray cans of isopropyl alcohol available on the market.
- C) Spray well on the sample-holder.
- D) Leave turned off for at least one hour before beginning a new work cycle or any other type of operation on the instrument.



Instrument may be damaged if dropped.

Shipping

Put the instrument into its plastics bag and place into its own box as shown in the installation guide.

If the instrument is being shipped to the manufacturer for repairs, affix a copy of the "Fault Report" form to the outside of the box.

2 SERVICE MENU

2.1 Service functions

2.1.1 Access to the service menu

Starting up the system

Procedure

1. Turn on the instrument.
2. When the instrument turns on, it shows the following menu:

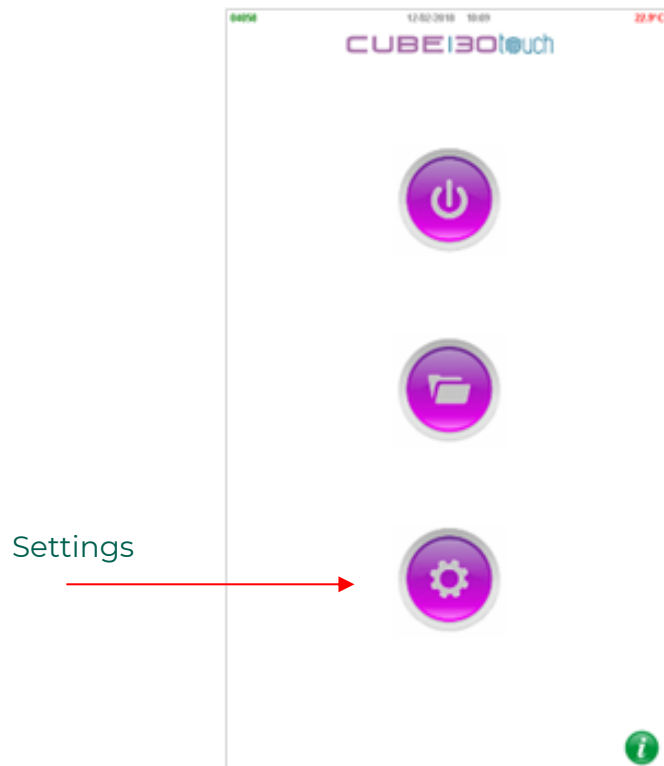


Figure 2.1

3. Press the Settings Button to access the Settings Menu
4. Press SERVICE to access to the ACCESS CODE Input Page:

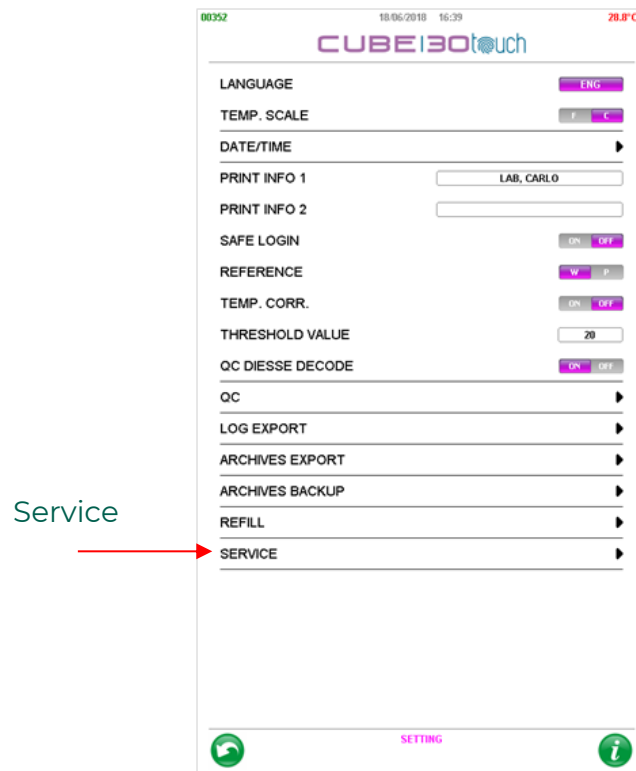


Figure 2.2

In the Top part of this page you can find the current **Firmware Version** and its release date.

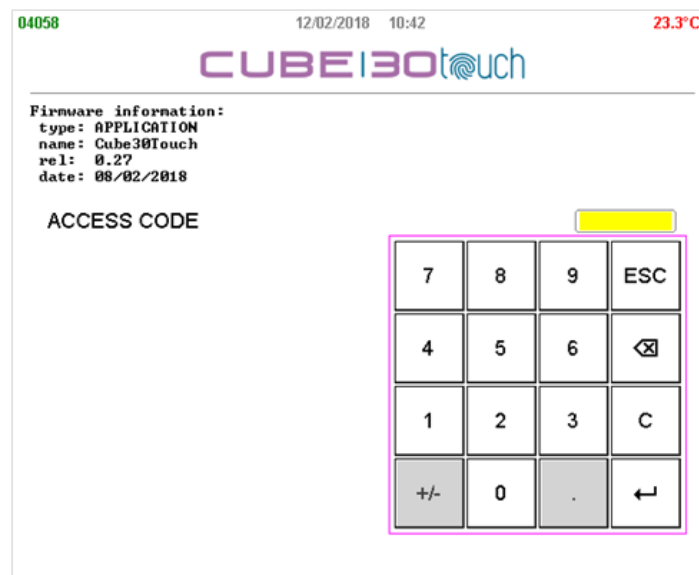


Figure 2.3

ACCESS CODE LIST:

98521	14785	74159
		
SERVICE	PARAMETERS	VIEWER

Other passwords:

'99991' Saving screens:

The instrument beeps to confirm the activation of the function.

To save the screens insert a pen drive and click on the "CUBE 30 touch" logo at the top, the instrument will emit a beep and start saving, after a few seconds it will emit a further beep to confirm the saving. If the procedure encounters problems, a lower tone beep will be emitted.

The .BMP files are saved in the "SSHOT" subfolder of the pen drive. Each SH0000x.BMP image will have dimensions (600x1024 pixels) (1.75MB).

'85209832' Fast Analysis in 2 Minutes:

No Mixing (4/5 revolutions), No Sedimentation - (For Host connection test etc.)

Switch Off instrument for return to normal cycle. ***Just for test!***

In case the **SafeLogin** is active in the instrument, and the Diesse technician must access the instrument, not having a disposition none of the registered users:

can do so by selecting the first one user of the list (the administrator) always present, and enter the password service preceded by a '0' character. (for example, **098521**)".

2.1.2 Service menu

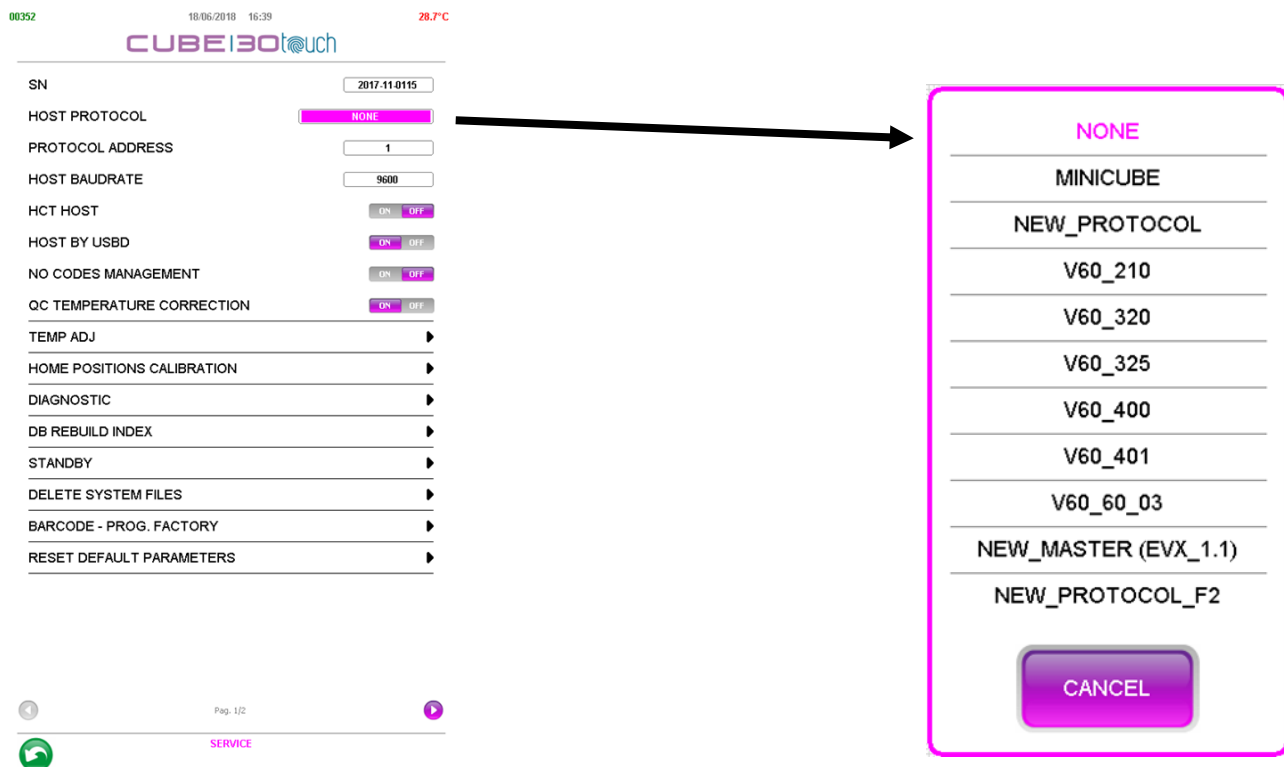


Figure 2.4

Page 1/2:

Function		Description	Default
SN	2017-11-0115	The instrument's serial number	
HOST PROTOCOL		You can choose between different types of protocols. Monodirectional (V60-xxx Type) and bidirectional. Some of these are present for compatibility with the "Original" Ves-Matic line instruments. Set address and baud rate parameters for any host protocol selected. - For the NEW_MASTER protocol: - if a sample is discarded from the host, the yellow error stamp containing the word "HR" (Host Reject) is displayed on both the plate and in the samples screen. - 500 mS of waiting ACK x 3 attempts - if the host responds to the ACK, the response is expected for a maximum time of 15 seconds. In the event of an ACK or ANSWER error: the sample is executed.	none

HCT HOST 	For future uses, always set to OFF	OFF
HOST BY USB 	In USB /Serial link to ves matic Protocol to host: ON = on USB (USB HOST) with FTDI (USB/Serial Adaptor) OFF = on CDC (USB DEVICE): direct USB connection to the PC, requires “STM32” USB Driver to be installed on the PC.	ON
NO CODE MANGEMENT 	ON = No barcode reading is performed during the test OFF = Samples and QC barcodes are read and used during the test.	OFF
QC TEMPERATURE CORRECTION 	ON = The Q.C tubes result is corrected in temperature following the Manley's nomogram. OFF = No correction in temperature for Q.C. tubes, while the Samples are always corrected.	ON
TEMP ADJ >	Temperature setting: Measure the temperature inside the instrument and insert the value in the label.	
HOME POSITION CALIBRATION >	In this test the selected Motor executes a complete movement and the Home offset can be adjusted to align it perfectly to its home position. This menu is very important to set the Home position of the Plate, Reading Unit and Mixer movements. <u>WARNING: These values must not be changed.</u> Only if a mechanical part has been replaced or if the movements are not centered, it may be necessary to recalibrate these positions.	
DIAGNOSTIC >	Diagnostics on movements, transponders, readers, barcode etc. For the detailed description of the tests please see the paragraph “Diagnostics Functions”.	
DB REBUILD INDEX >	Rebuilt of the archives indexes. It may be required if some settings (in data/mode etc.) were changed or when the result list in the archives is not sorted correctly.	
STANDBY >	To set the screen save function and the brightness of the CUBE 30 TOUCH written. TIME STANDBY: Time of inactivity, in Seconds, after which the Instrument goes in Stand-by mode. Set= 0 to deactivate the Standby-by management.	900
	TEST: to check the Stand-by Mode.	

DELETE SYSTEM FILES >	Allow to delete the files stored into the uSD Card in order to free the space on it. After deleting, all BMP images must be reloading from SYSTEM folder in USB stick at power up.	
BARCODE - PROG. FACTORY >	Send to the Barcode Reader the factory configuration. To be used once a new reader is installed.	
RESET DEFAULT PARAMETERS >	Resets all the parameters, including the mechanical calibrations to their default values. WARNING: After this action the Motors Offsets and Temp. Adjust. Must be performed by the command: "HOME POSITION CALIBRATION" and "TEMP_ADJ". Perform Only if a mechanical part has been replaced.	

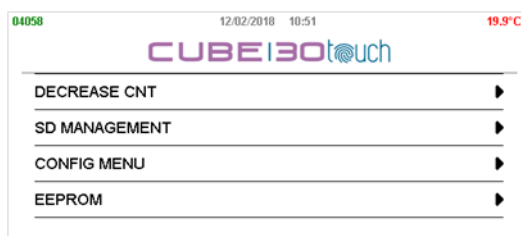


Figure 2.5

Page 2/2:

Function	Description
DECREASE CNT >	You can decrease the number of available TESTS. CAUTION: EXAMS (TESTS) removed cannot be restored.
SD MANAGEMENT >	HELP IMPORT: Import from a USB key all the .bmp files and help_xxx.txt. NOTE: This function is also imports the image of initial presentation "hello.bmp" HELP DELETE: Delete all help_xxx.txt and .bmp files. NOTE: it also erases the image of initial presentation "hello.bmp" ARCHIVES DELETE: Delete all archives LOG DELETE: Delete all log files SD BACKUP: Copy all files to internal CUBE 30 TOUCH in a USB key It is advisable to do this in a completely empty key SD RESTORE: Copy all the files in a USB key in uSD inside the CUBE 30 TOUCH SD FORMAT: Formats the internal uSD to CUBE 30 TOUCH.

	WARNING: This operation involves the loss of all the internal files to CUBE 30 TOUCH.
CONFIG MENU >	EXPORT: Export the Configuration Parameters into a file on a USB Pen Drive PRINT: Prints the list of the Configuration Parameters In this print out of the configuration are included the Motor Home Position setting! HOME STEP Plate, Reader and mixer. DEFAULT: Set the configuration Parameters to their default values
EEPROM >	EXPORT: Export the binary image of the EEPROM memory on a USB Pen Drive, the file name will be: nnnn_EEP.BIN, where nnnn is the serial number of the instrument. IMPORT: Imports a binary image of the EEPROM from a file located on a Pen Drive. The file will be imported only if its name includes the same Serial Number of the instrument (see EXPORT function). WARNING: The Import Operation will overwrite all the calibrations, including Motor Home Position and Temperature Setting!!

2.1.3 Home position calibration

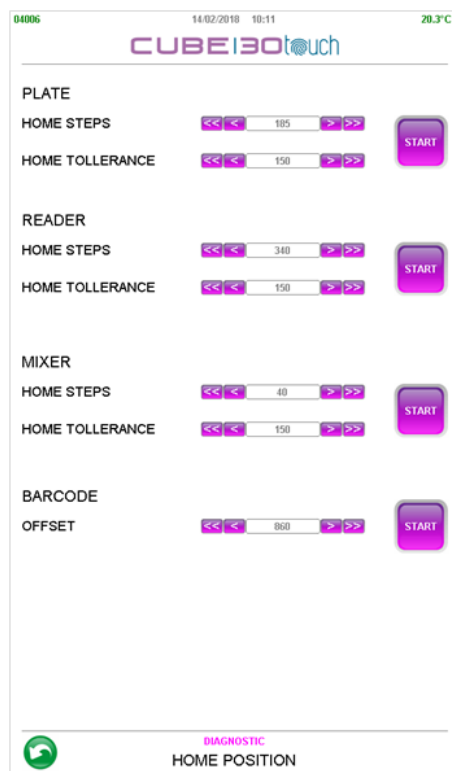


Figure 2.6

In this test the selected Motor executes a complete movement and the Home offset can be adjusted to align it perfectly to its home position.

This menu is very important to set the Home position of the Reading Unit.

If the Motor or the Home Optical Sensor have been dismantled or replaced a new Home Position Setting is needed.

Once the correct positioning has been reached press START, to verify the new Value, otherwise adjust the Home Steps value, and repeat the test pressing Start.

To adjust the **Sample Plate Home positioning**, that is the most critical one, you have to put two sample's tubes in the two reading positions 1 and 2 of the Sample's Plate.

Start the procedure by pressing the Start button related to the Reader.

The Plate will move to its home position, then the Reader will move up and stop at the topmost position for a few seconds then returns back to its home position. If the reader is not perfectly centered respect the two sample's tubes, adjust the Plate Home Steps value and repeat the procedure.

Adjust the **Mixer Home positioning** by in order to place the Plate in a perfectly horizontal position, without hit the metallic support plate. Adjust the Mixer's Home Steps value and press Start to check its alignment.

Adjust the **Reader Home Positioning** in order to have it lower than the longest sample's tube and without hitting the metallic support plate.

2.1.4 Diagnostics functions



Figure 2.7

2.1.4.1 Touch panel calibration (touch screen calibration)

To enter the Touch panel calibration proceed as follow:

1. Power off the Unit
2. Press the Touch panel and power on the unit keeping the Touch panel pressed, until the Touch Calibration page appears on the display.
3. Release the Touch panel to enter in the Calibration page.
4. Press the center of the small circles draw using a pen at the end of the procedure the new values will be saved.

2.1.4.2 Buzzer test

Press Buzzer from the Diagnostics menu and hear the sounds generated by the Buzzer.

2.1.4.3 Eeprom test

Press EEPROM to enter in the EEPROM test page, then press START to execute the test procedure.

At the end of the procedure the initial content of the EEPROM memory is restored.

WARNING: if the procedure fails or the power supply is removed during this test the EEPROM content will be lost! Pay attention to export the EEPROM on a USB PENDRIVE before to start this test.

2.1.4.4 SD card test

The SD CARD (in the CPU board) contains the images used by the software and also the archives and log files. Press SD CARD to enter in the internal SD CARD test page, then press START to execute the test procedure.

If it is completed successfully, the SD CARD is working properly. During this procedure a test file is created and then deleted from the SD CARD, all the other files are kept unchanged.

2.1.4.5 Transponder test

Press TRANSPONDER to enter in the internal TRANSPONDER test page, then press START to execute the test procedure.

The test is made in two phases: first the RF system is tested by measuring the frequency of the RF carrier that has to be 125KHz +/- 3KHz; then the Check Device Transponder is checked by reading the number of Exams it stores to be loaded in the Unit.

2.1.4.6 Home sensor test

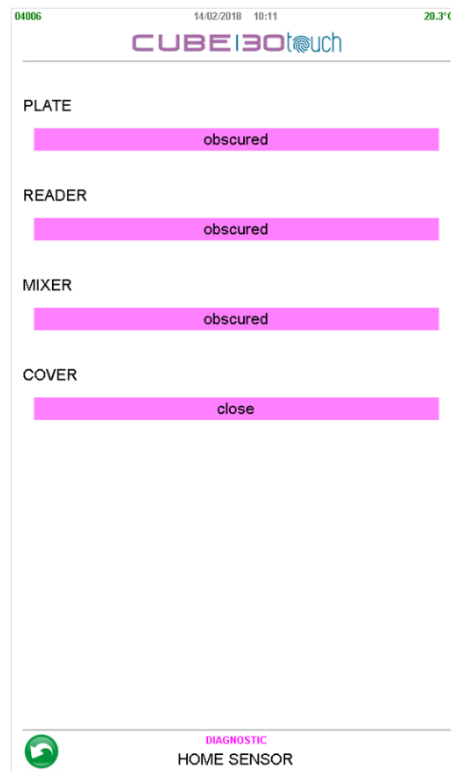


Figure 2.8

This test report the status of the Home Sensor of the three Motors and the status of the caver microswitch that could be:

- **Obscured** (the carriage is in its Home position)
- **Free**

2.1.4.7 Motor test

In this section the Plate Motor is moved Clockwise ad counterclockwise by 30000 steps.

2.1.4.8 Sensor tube test

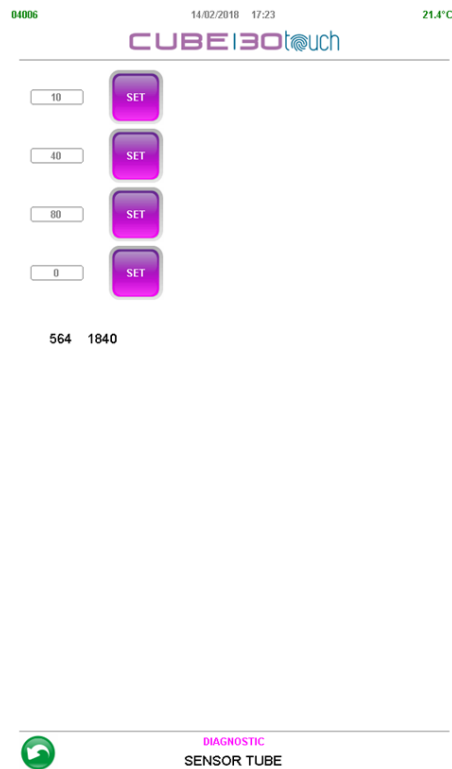


Figure 2.9

By entering in this menu it is possible to set the light levels of the 2 white LEDs emitters (Range 0-100), and read the light level read by the corresponding Light Sensors.

LEDS TEST:

There are four predefined light levels written in the fields listed on the upper left side. They can be changed by touching them. By pressing the purple button near the light level field, the LEDs will be driven to emit the corresponding light. You can check the light levels for the two LEDS at the same time.

DARK TEST:

In the bottom part you can read the Sensors light readings.

When all the LEDS are OFF (pressing the button near the "0" level) and by closing the upper cover. You have to check that the Light Read by the Sensors is less than 200.

2.1.4.9 Barcode (internal) test

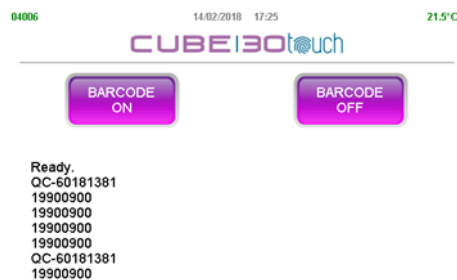


Figure 2.10

Press the Barcode ON button to activate the Barcode Reader, then place any barcode label in front of it to check if it's properly read and printed on the screen.

It is also possible to insert sample's tubes with labels in the sample plate and manually rotate it in order to test if their barcodes can be read by the reader.

Press Barcode OFF before exit.

Model type: LEUZE Electronic CR50

From Barcode reader USER MANUAL, the default settings:

UPC/EAN

- UPC-A Enabled
- UPC-E Enabled
- EAN-13 Enabled
- EAN-8 Enabled
- Bookland EAN(ISBN/ISSN) Disabled
- Addendum Disabled
- UPC-E to UPC-A format Disabled
- UPC-A to EAN-13 format Disabled
- Transmit UPC-A check digit Enabled
- Transmit UPC-A leading digit Enabled
- UPC-A data redundant check Off
- Transmit UPC-E check digit Enabled
- Transmit UPC-E leading digit Enabled
- UPC-E data redundant check Off
- Transmit EAN-13 check digit Enabled
- EAN-13 data redundant check Off
- Transmit EAN-8 check digit Enabled
- EAN-8 data redundant check Off

Code 39

- Code 39 Enabled
- Transmit Code 39 Start/Stop character Disabled
- Convert Code 39 to Code 32 Disabled
- Set Lengths for Code 39 4-30
- check digit verification Disabled
- Code 39 FULL ASCII conversion Disabled
- Code 39 data redundant check Off

Code 93

Code 93 Enabled
Set Lengths for Code 93 4-30
check digit verification Disabled
Code 93 data redundant check Off

Code 128

Code 128 Enabled
UCC/EAN-128 Enabled
Set Lengths for Code 128 4-30
Code 128 FNC2 concatenation Disabled
Transmit UCC/EAN-128 FNC1 character Disabled
Transmit Check digit Disabled
Code 128 data redundant check Off

Codabar

Codabar Enabled
Set Lengths for Codabar 6-30
Codabar Start/Stop character A,B,C,D
check digit verification Disabled
Codabar data redundant check Off

Interleave 2 of 5

Interleave 2 of 5 Enabled
Set Lengths for Interleave 2 of 5 10
check digit verification Disabled
Interleave 2 of 5 data redundant check 1

Code 11

Code 11 Disabled
Set Lengths for Code 11 6-62
check digit verification One check digit
Transmit check digit Disabled

Standard 2 of 5

Standard 2 of 5 Disabled
Set Lengths for Standard 2 of 5 6-62
check digit verification Disabled

Matrix 2 of 5

Matrix 2 of 5 Disabled
IATA Code Disabled
Set Lengths for Matrix 2 of 5 6-62
check digit verification Disabled

Industrial 2 of 5

Industrial 2 of 5 Disabled
Set Lengths for Industrial 2 of 5 6-62
check digit verification Disabled

Chinese Postal Code

Chinese Postal Code Disabled
Set Lengths for Chinese Postal code 10-16
Chinese Postal code data redundant check 1

MSI/PLESSEY

MSI/PLESSEY Disabled
Set Lengths for MSI/PLESSEY 6-30
check digit verification Double digit check
Transmit check digit Disabled
MSI/PLESSEY data redundant check 1

GS1 DataBar

GS1 DataBar Omnidirectional Disabled
GS1 DataBar Limited Disabled
GS1 DataBar Expanded Disabled
Transmit GS1DataBar Omnid. check digit Enabled
Transmit GS1 DataBar Omnid. Application ID(01) Enabled
GS1 DataBar Omnid. to EAN-128 emulation Enabled
Transmit GS1 DataBar Limited application ID(01) Enabled
Transmit GS1 DataBar Expanded application ID(01) Disabled

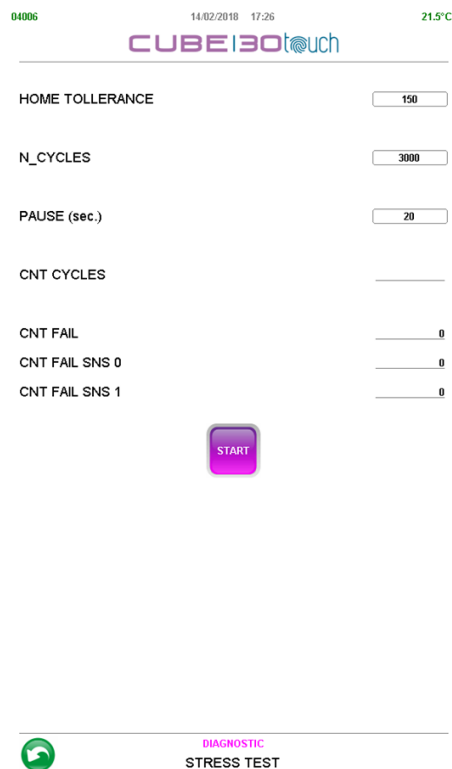
Telepen

Telepen Disabled
AIM Character Telepan Enabled

ADDITIONAL SETTINGS from FIRMWARE CUBE30T (BARCODE - PROG. FACTORY)

- Codabar minimum length setting: 1
- Codabar maximum length setting: 30
- ITF 2 of 5 code minimum length setting: 1
- ITF 2 of 5 code maximum length setting: 30
- Code 128 minimum length setting: 1
- Code 128 maximum length setting: 30
- Codabar start/stop character transmission = None

2.1.4.10 Stress test



04006 14/02/2018 17:26 21.5°C

CUBE130touch

HOME TOLLERANCE 150

N_CYCLES 3000

PAUSE (sec.) 20

CNT CYCLES

CNT FAIL 0

CNT FAIL SNS 0 0

CNT FAIL SNS 1 0

START

DIAGNOSTIC
STRESS TEST

Figure 2.11

Use this function to stress all the Motors and the Reading Unit.

Put some samples tube and close the cover.

Press Start once you have set the parameters.

The test will perform One Mixing + Plate rotation + Reading Procedure for each cycle.

A step loss control and the light level of each sensor is monitored cycle-by-cycle. An error counter for the Motor's error is increased for each step loss error.

And an error counter for each sensor is also incremented if the light level is outside the expected range (dark when the corresponding LED is OFF; light when the Led is ON).

This function can be useful to check the integrity of the unit's hardware.

Normally the suggested number of cycles is 200 maximum, with a pause of 20 seconds between the cycles.

2.1.5 Reading parameters

In this menu the parameters used to calculate the blood level and the ESR values are accessibles and may be varied.

Pay attention : changing these values may affect the performances of the Instrument.

The values below are the default values actually used on the unit.

NUM_READINGS	2	Number of Readings (1 st after mixing, 2 nd after 20 Min.)
DETECTION_STEP	2200	(<i>Starting from SW.1.12</i>) Number of steps of the readers (Up movement) for the sample detection phase (cycle start). 2200 is about halfway between home position (on the bottom) and the top (under cap). To climb up to the plate springs, the value is 3500 (for detect pediatric tube put around 3200).
SED.TIME	1200	Time between Readings in seconds
MIXING CYCLES	20	Number of complete rotations for samples mixing
LIGHT_STANDBY	1	Power led in % - Used when there is no tube
LIGHT_DETECT	1	Power led in % Used for detect the presence of tubes
LIGHT_NORMAL	10	Power led in % For scan normal tubes (1 st and 2 nd readings)
LIGHT_PEDIATRIC	1	For scan pediatric tubes (Parameter not used in Soft. rel. 1.xx)
LIGHT_TEST_EXTRACT	0	Parameter not used

M1 is the mathematical method used for detecting the BLOOD Level (BLD. L.) (n° of steps) in the graph.

M1.AVG_N_SAMP_PRE	2	Filtering with moving average.
M1.AVG_N_SAMP_POST	2	A new curve is drawn where each point is the average of +/- 2 points. (Pre/Post processing filter values).
M1.N_RISING_BOTTOM	15	For the research of the tube bottom: starting from right to left, we search for 15 consecutive steps in which the value of light decreases. The first one is taken.
M1.T_BOTTOM_LIM_MIN	500	Tube Bottom minimum position acceptable
M1.T_BOTTOM_LIM_MAX	950	Tube bottom Maximum position acceptable
M1.T.BOT_PED_LIM_MIN	200	Tube Bottom minimum position for pediatrics samples (<i>not in use</i>)
M1.T.BOT_PED_LIM_MAX	500	Tube Bottom Maximum position for pediatrics samples (<i>not in use</i>)
M1.SH_TRESHOLD	1500	Sample High Threshold: this is a light level in digits of the A/D converter. If in the top half of the tube the light level never goes over this limit, it is assumed that the blood level is too high inside the tube, and that cannot be determined because it is over the upmost position reachable by the reading sensor. A SAMPLE HIGH error is issued for this sample. <u>Notice that this condition could happen also if the number of labels attached to the sample is too high, avoiding the light to pass through.</u>
M1.PERC_FILTER	15	The method M1 to identify the point (steps n°) in which the blood level begins (or sedimentation), uses the "filtered" graph. The filtered graph is the same as the original graph but when the light value is less than M1.PERC_FILTER, it is set equal to this value.
M1.DIFF_DIGITS	100	Parameters used to find the point (blood level in the 1 st or sedimentation level in the 2 nd .)
M1.DIGITS_RATE	200	
M1.N_FALLING	12	The graph of the readings from left to right is analyzed. Stop at the first point corresponding to the filtered light value. Go back to (12) steps, checking for a progressive increase in the light value and the intermediate point (sixth) is taken.

M1.SL_THRESHOLD	245	SAMPLE LOW threshold. If in the first reading the blood level measured respect to the Tube bottom (difference) is less than this value, the sample is marked as SAMPLE LOW . This value is the difference: Bottom level - Blood level (BOT.L. – BLD.L.).
-----------------	-----	--

The Control Method (CM) used is the M2 Method which studies the trend of the DERIVATIVE graph.

The CM finds the point of blood level. If this value is similar to that found by M1.

(in an acceptable range = CM. ACCEPTABILITY) the value of M1 is confirmed (otherwise error “-14” is given).

This comparison is made in each curve (first reading and the second reading).

CM.AVG_N_SAMP_PRE	5	Filtering with moving average for the Control Method .
CM.AVG_N_SAMP_POST	5	A new curve is drawn where each point is the average of +/- 5 points. (Pre/Post processing filter values).
CM.ACCEPTABILITY	60	Maximum difference admitted in the blood level detection (or in the sedimentation level detection) between Method 1 (M1) and Control Method (CM), if the difference is above this limit, the sample is marked with ERR. (-14)
CM.WIN_BOTTOM_DIST	50	The study of the derivative graph (CM) stops 50 steps before the tube bottom.
CM.MIN_ANG_COEFF	25	Minimum value of the angular coefficient (in the derivative graph) which determines a sensitive point.
CM.RISING_EDGE_DIST	0	If a positive derivative of equal value is found within the next 45 readings, the point previously found is canceled and a new one is sought. <i>Starting from Software Rel. 1.13 if Parameters</i> CM.RISING_EDGE_DIST = 0 only <i>negative descendant derivatives below CM.minAngCoeff are taken.</i>

VES_ADJ	1000	
VES_ADJ_PED	1000	
VES_ADJ_QC_DIESSE_N	1000	
VES_ADJ_QC_DIESSE_A	1575	
VES_ADJ_QC_CUSTOM_N	1000	

VES_ADJ_QC_CUSTOM_A	1000	
ESR_ZERO_ENABLED	1	If equal to 1 a result equal to zero can be obtained.

BL_HEIGH_3000_UL	580	Next use - Parameters for determining the exact volume of blood in the tube.
BL_HEIGH_1500_UL	240	

Calculation curves:

N_PNT_CURVE	6			
DELTA.1	1		ESR.1	0
DELTA.2	50		ESR.2	13
DELTA.3	150		ESR.3	40
DELTA.4	240		ESR.4	90
DELTA.5	290		ESR.5	120
DELTA.6	180		ESR.6	172

----- not in use -----

N_PNT_CURVE_PED	5			
DELTA.1	5		ESR.1	1
DELTA.2	30		ESR.2	15
DELTA.3	70		ESR.3	30
DELTA.4	90		ESR.4	60
DELTA.5	140		ESR.5	60

N_PNT_CURVE_VOL_ADJ	4			
DELTA.1	1500		ESR.1	135
DELTA.2	2000		ESR.2	120
DELTA.3	3000		ESR.3	1000
DELTA.4	4000		ESR.4	80

2.1.6 Reading viewer (password: 74159)



Figure 2.12

This part of the program displays all graphs of the samples present in the LOG.txt file. As you enter it's showing the latest LOG. File reading. The graphs displayed are those of the last sample of the last cycle.

Below the date of the log.txt file (a file log, covers a day), with the right and left arrow keys to scroll forward and backward through the log files. Below the time of the test. If the readings are two (full examination) this time of the first and the second reading. Below there is a window with the bottom two buttons to scroll through two different pages.

In the first page there is information of the loaded reading.

BOT.L	= Bottom level
BLD.L	= Blood level
ERR	= The number of Error is showed if present
DELTA	= Delta between readings (for comprehensive examinations only)
ESR [W]	= ESR calculated (for comprehensive examinations only)

R1	= Column of values in the first reading log
R2	= Column of values in the second reading log
R1E	= Column of the values of the first reading elaborated with the current parameters of the readviewer
R2E	= Column of the values of the second reading revised with the current parameters of the readviewer

ESR is always presented in Westergren and cannot take account of two corrections to HCT because while saving the log, the HCT is not present.

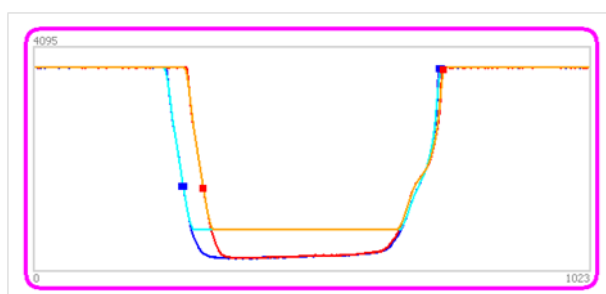


Figure 2.13

Note: The graph shown with different colors the tracks that originated the date of four columns

Blue = R1
Red = R2
Cyan = R1E
Orange = R2E

The graph marked with blue squares (for the first reading) and red (for the second) points identified bottom level and blood level.

Note:

The temperature is only stored in the second reading. In case of single read the temperature is not displayed. It is Celsius degrees and it is green or red depending on whether the measure had or not the active correction.

The Second Page shows the parameters used for processing.

N.RISING	25	SH_TSH	1500
PER FIL	15	DIFF DIG	100
DIG RAT	200	N.FALL.	12
SL TSH	245	T.CORR	ON
CTRL METHOD	ON		
AVG_N_SAMP_PRE	5	AVG_N_SAMP_POST	5
DELTA TO ACCEPT			35
WIN_BOTTOM_DIST	50		
MIN_ANG_COEF	35		
RISING_EDGE_DIST	45		
EDGE FOUND			
		268	308

Figure 2.14

In this page each parameter can be changed. After each change the data and the graphs are upgraded.

The Temp Corr. ON / OFF setting only affects the calculation of the ESR and is only active if there are both readings. The changes only affect the ReadViewer session. Exiting and reentering the ReadViewer restores in the ReadView all the values present in the instrument settings.

Also, it shows the parameters related to the control method.

The last two numbers displayed in the boxes at the bottom right show the points identified by the control method (CM) in the first and second reading.

Notice that if modifying the various parameters, the blood level is not detected (for the CM method) the ERR -13 is displayed, while if the level is found, but the two methods find a different point, if the difference is bigger than the CM. ACCEPTABILITY parameter, the ERR -14 will be displayed.

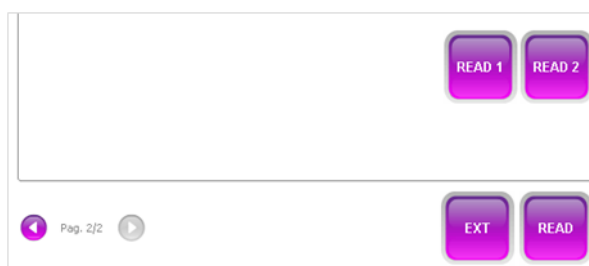


Figure 2.15

The command “**READ1**” or “**READ2**” allows the reading of a tube, in real time.

It’s possible to choose the power of the light, for each one of the two sensors.

The command “**EXT**” allows to check the LOG files presents in an external USB KEY.

Starting from software version SW. 1.11 Parameters are also exported to the LOG.txt file.

Also, from this version are the values found by method 2 (M2=CM) (now they are visible in the PC software “**Read Viewer 1.04**”).

2.1.7 Error table

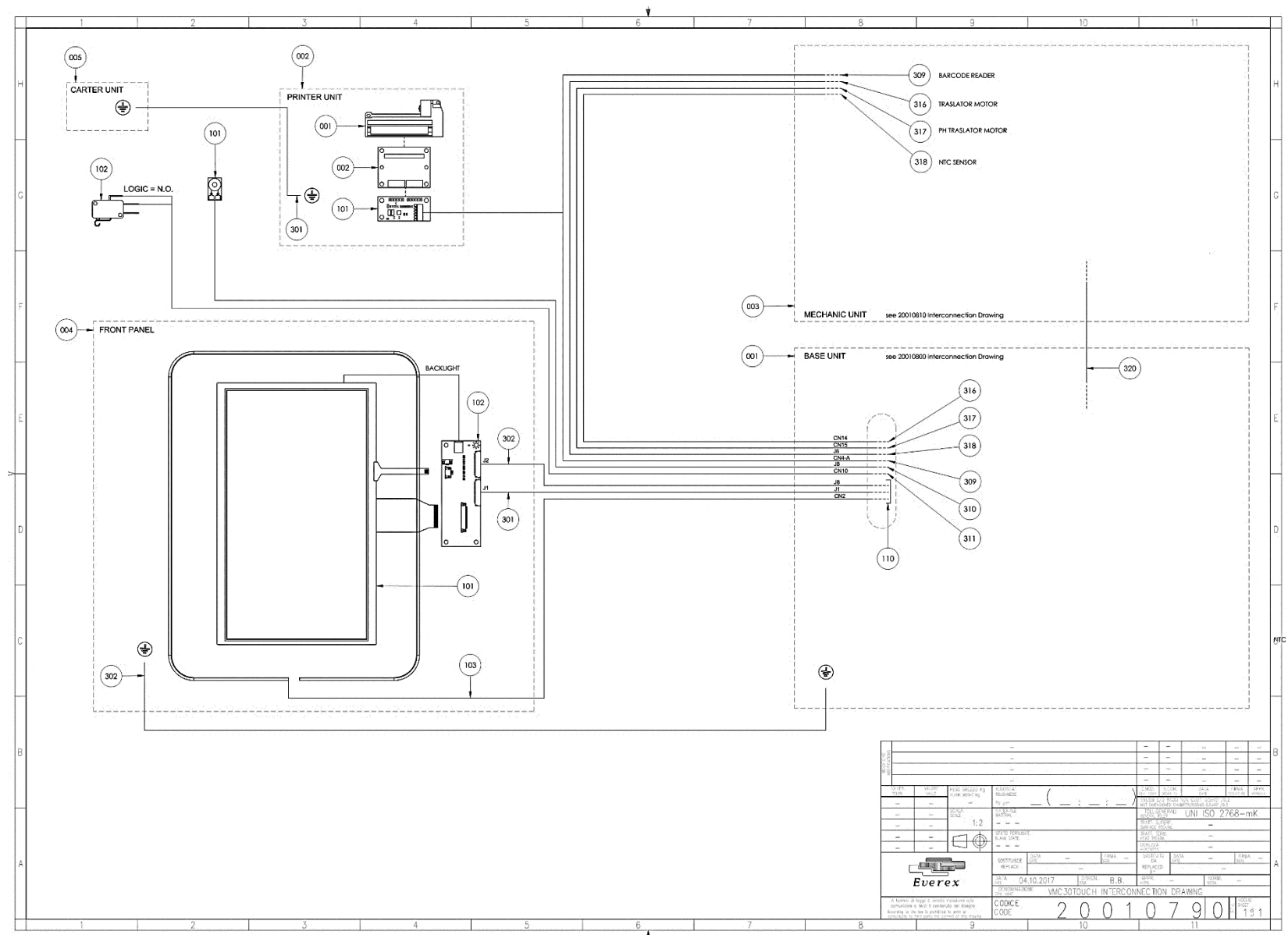
CUBE 30 TOUCH - ERROR TABLE			
1	ERR	Generic error	
2	ERR	Tube Bottom not found	
3	ERR	Tube bottom is out of the pediatric and normal range	
4	ERR	SH_TSH not found	
5	HIGH	SH_TSH found too low (more than half the normal tube)	
6	ERR	BLOOD LEVEL not found	
7	LOW	BLOOD LEVEL found over SL_TSH level	
8	ERR	BLOOD LEVEL second reading beyond the first one	
10	>140	(ESR result > 140)	
11	ERR	Tube removed	
13	ERR	Control method CM failed	
14	ERR	Failure to confirm the control method CM	
The -10 is not an error but denotes that the value of ESR is above 140 and is displayed with "> 140"			

3 LEARNING ABOUT THE UNIT- DESCRIPTION OF THE MAIN MODULES

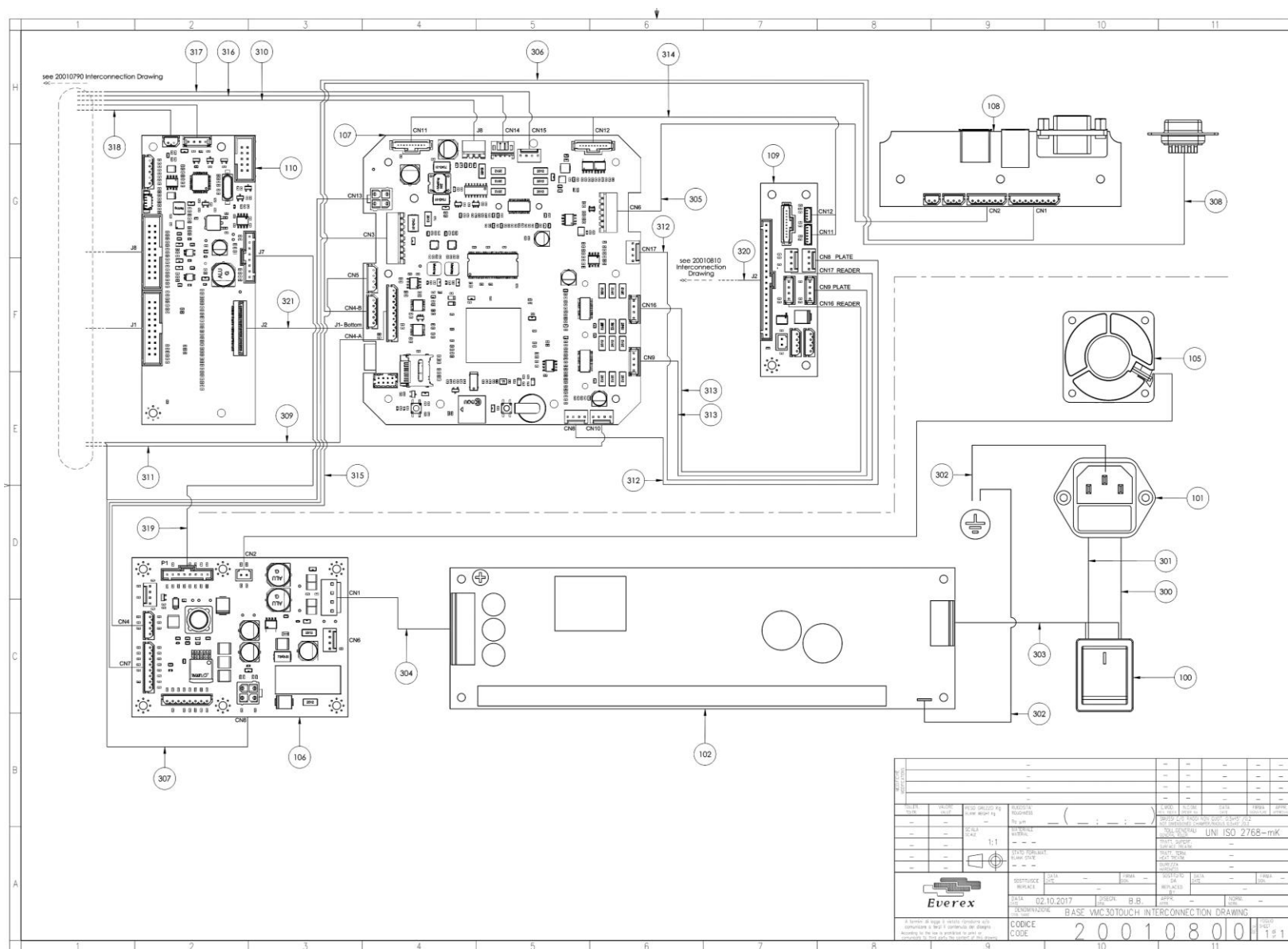
3.1 Interconnection diagrams

See the Parts List for the reference numbers

3.1.1 General interconnection drawing



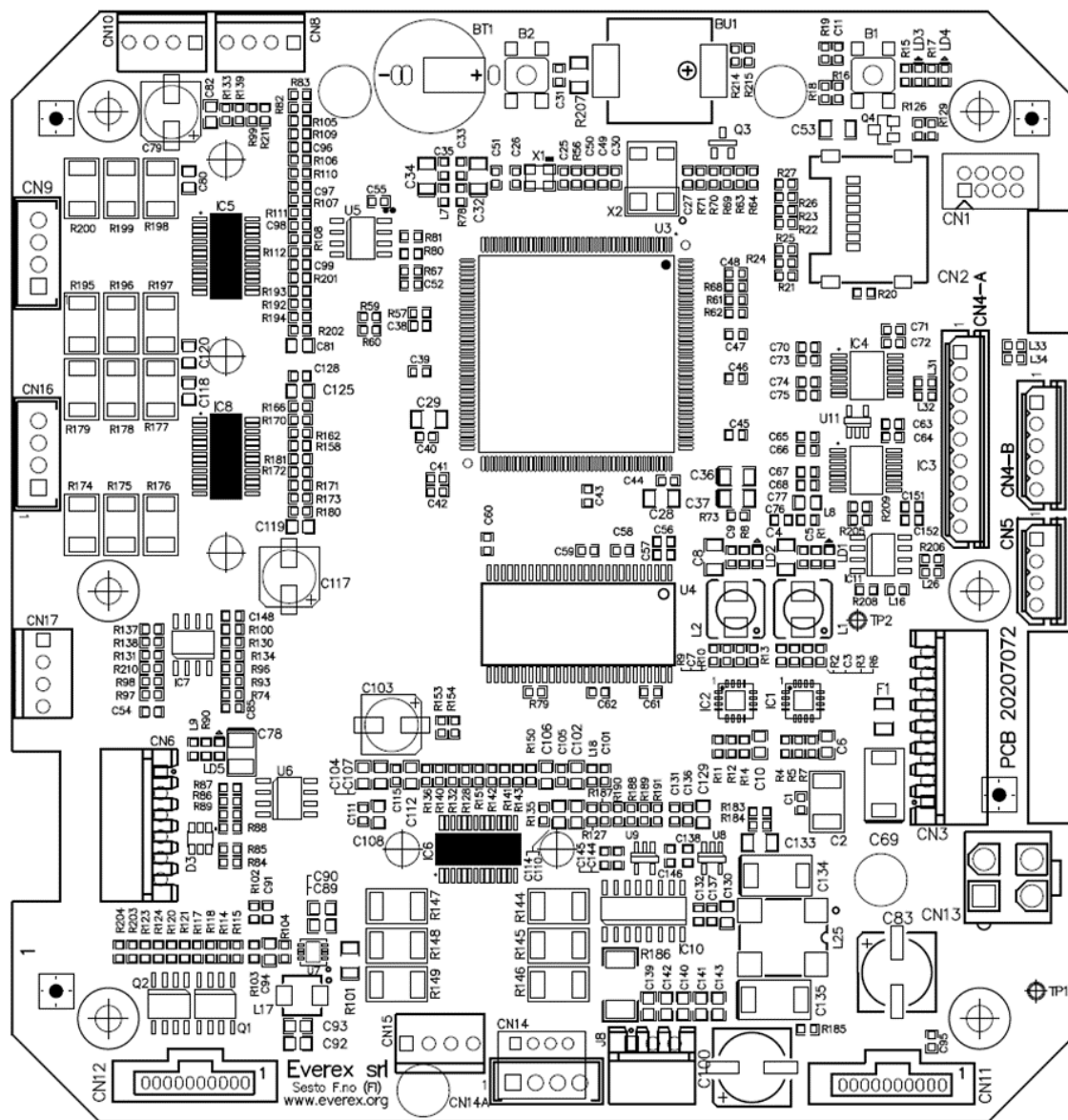
3.1.2 Base ass.y interconnection diagram



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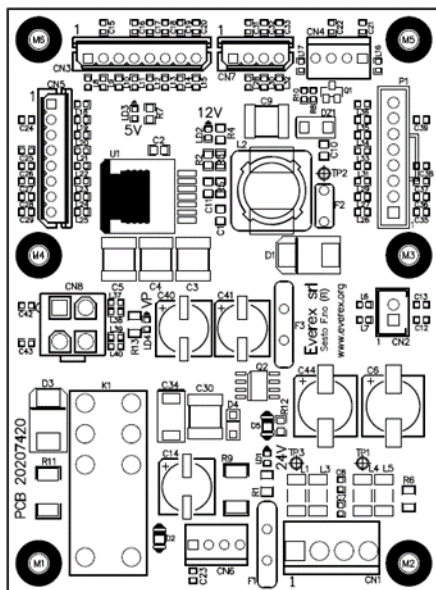
3.2 CPU board connections



30136901 - CPU Cortex M4 180MHz TFT + 3 Step Mot Board

- CN14: Stepping Motor
- CN15: Home Sensor Optical Switch
- CN12: Reading Sensors LEDs
- CN11: Reading Sensors Light Detectors
- CN6: USB Ports, to be connected to the Interconnection Board in the Base block
- CN3: 12Vdc Power Supply + Barcode Reader RS232 Port + RF Antenna output for the transponder reader. To be connected to the Interconnection Board in the Base block.

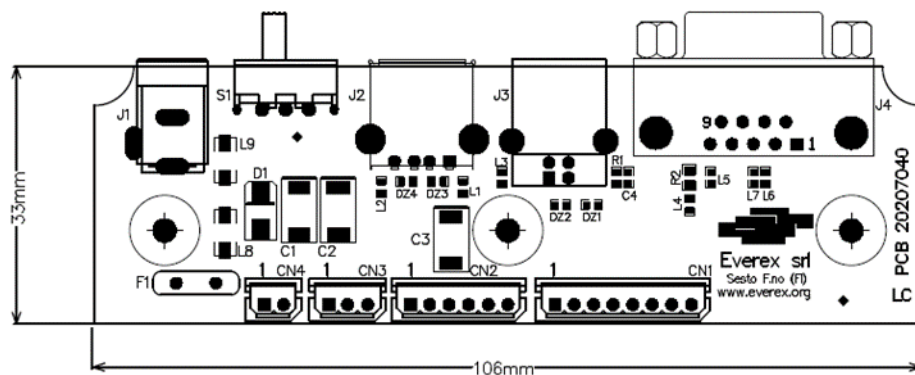
3.3 Power supply interconnection board connections



30137930 – Power supply Interconnection Board

- CN1: 24V input
- CN2: FAN power supply
- CN6: LID Microswitch in
- CN8: Motor Power Supply Voltage (24V) for CPU (CN13)
- P1: Control Display Board Bus connection (P1)
- CN7: RS485 in from CPU (CN3)
- CN5: External Interconnection Board (CN1)
- CN3: 12V Power supply + RS232 connection for CPU (CN5)
- CN10: Lid Sensor OUTPUT to CPU Board (CN10)

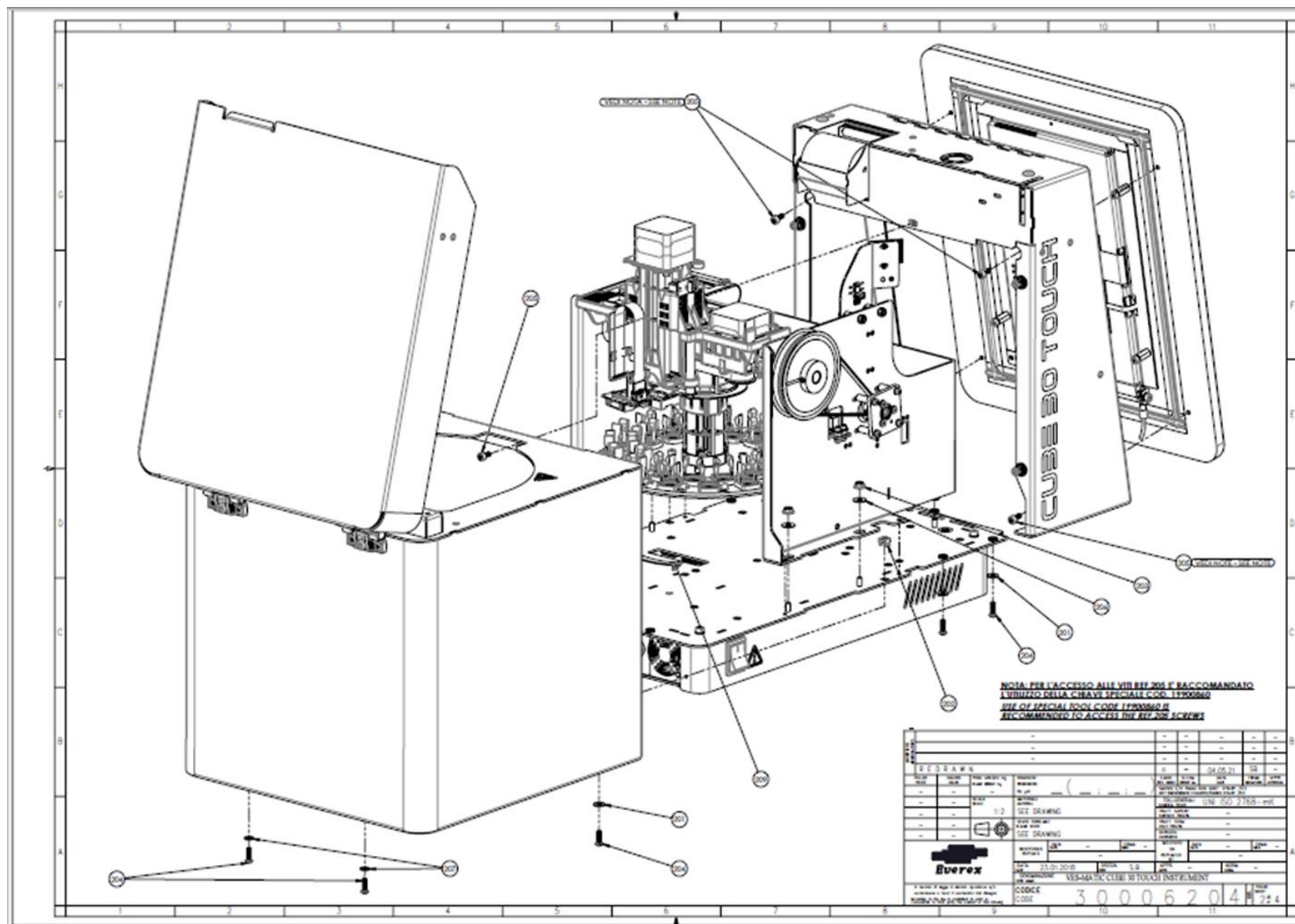
3.4 External interconnection board

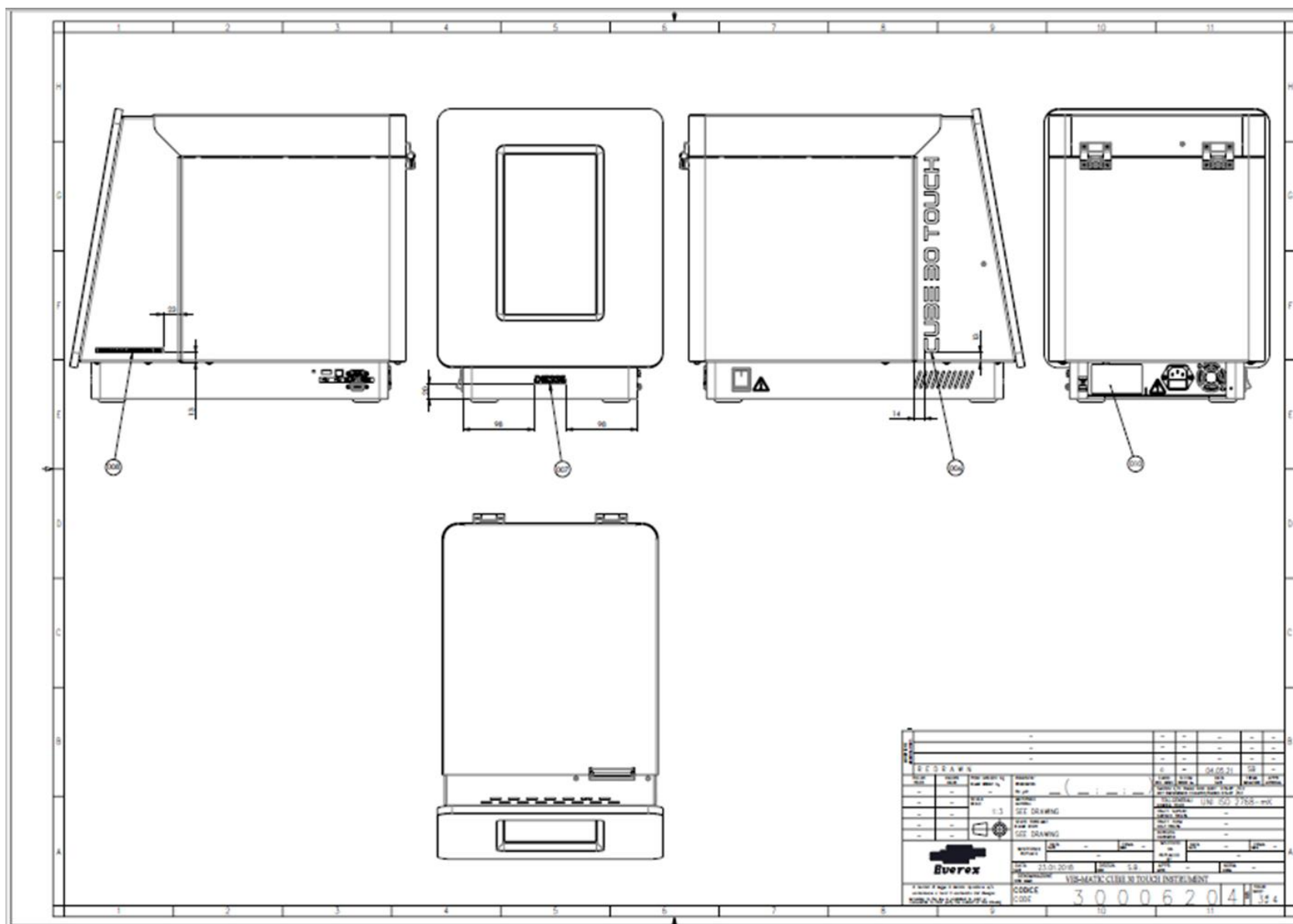


30137980 – External Interconnection Board

- J2: USB HOST port
- J3: USB DEVICE port
- J4: Barcode RS232 port
- CN3: Transponder Antenna
- CN2: USB PORTS input from CPU board
- CN1: POWER Supply out for the CPU Board + RS232 signals for the Barcode reader and the RF Antenna.

[illegible]





P/N	Description	Q.ta	Reference
10807931	VES MATIC CUBE 30 TOUCH (MET) MATRICULAR LABEL	1	010
10809740	LOGO DIESSE 43,6x11 PRE-SPACED ADHESIVE LABEL	1	7
10809750	"CUBE 30 TOUCH" PRE-SPACED ADHESIVE LABEL	1	6
10809760	"Designed and assembled in Italy" LABEL DIM. 92,8x6,3	1	8
1110009Z	WASHER STEAB 5015/4/1/16 - D10 d4,3 H1	6	201
1110H44Z	CAGE NUT M4 INOX (CLIP FOR THICK.2mm)	2	202
1110L25Z	M4 0394 004 WHITE STAINLESS STEEL HEX FLANGE NUT	4	203
1110L75Z	SCREW TBEI M4x12 ISO7380 INOX A2	8	204
11141384	SCREW M4X8 UNI 5931 INOX	5	205
1114474L	HEX. NUT M4 UNI 5588 INOX	1	208
1114544Z	WASHER 4X12 INOX	4	206
1114B62Z	SERRATED LOCK WASHER A 4,3 DIN 6798 INOX	4	207
20920630	CABLE HOLDER ESSENTA FCCS-5	1	209
30221361	BASEMENT ASS'Y	1	001
30221391	INTERNAL MECHANICS ASS'Y	1	003
30222091	VES-MATIC CUBE 30 TOUCH FRONT PANEL ASS'Y	1	004
30222400	CUBE 30 TOUCH CARTER ASS'Y	1	005
30222480	SHIPPING LOCK ASS'Y	1	011
30222940	CUBE 30 TOUCH FIXED CARTER ASS'Y	1	002
30508340	CUBE 30 TOUCH ACCESSORIES KIT	1	

NOTE: for the actually orderable part numbers refer to the official spare parts list

P/N	Description	Q.ta	Reference
10342600	WASHER	4	9
103A4085	BASEMENT ASS'Y RAL 7012	1	1
103A4222	BASEMENT BOTTOM RAL 7012	1	2
10803350	"ELECTRIC SHOCK" LABEL 25x25 UL	1	4
10803360	"CAUTION" LABEL 25x25 UL	1	5
10803370	WEEE LABEL 15x20 UL	1	6
10803380	GROUND LABEL 13x13 UL	2	7
10807850	VMC30T EXTERNAL CONNECTIONS LABEL	1	8
1110B76Z	ROSETTA DENT.CONICA 4,3 DIN6798V ZN	1	201
1110M60Z	SCHOCK/VIBR. ABSORBER FOOT ESSENTA SVF-1 (TPE, 75ShA)	4	205
1110M79Z	HEX. SPACER F-F M4x45 ZINC-PLATED STEEL (RIF. WE 970 450 471)	4	204
11141182	SCREW M3X8 UNI 7687 INOX A2	22	206
11141193	SCREW M3X10 UNI 7688 INOX	6	207
11141242	VITE M3X20 UNI 7687 INOX	2	208
11141383	SCREW M4X8 UNI 7688 INOX	4	209
11141403	SCREW M4X12 UNI 7688 INOX	4	220
11141424	SCREW M4X16 UNI 5931 INOX	4	210
1114473L	HEX. NUT M3 UNI 5588 INOX	2	212
1114474L	HEX. NUT M4 UNI 5588 INOX	5	213
1114499Z	WASHER 3,2X7 UNI 6592 INOX A2	2	219
1114549Z	WASHER 6X18 INOX	4	214
1114B36Z	WASHER A 3,2 UNI 1751 GROVER INOX	2	218
1114B37Z	WASHER A 4,3 UNI 1751 GROVER INOX	4	211
1114B61Z	SERRATED LOCK WASHER A 3,2 DIN 6798 INOX	2	215
1114B62Z	SERRATED LOCK WASHER A 4,3 DIN 6798 INOX	6	216
20400500	FUSE 10A DELAYED TYPE 5X20 GT520310 [OMEGA] UL	2	104
20814070	DB CONN. FIXING KIT M/F NIKEL L=7	1	217
21200540	SWITCH BIP. LIGHT C. R5-1-5-K-K-F-T-O-F	1	100
21201080	FILTER FOR P.SUPPLY SCHAFFNER FN9260-10-06 UL	1	101
21445000	SW.POWER SUPPLY MEAN-WELL LPS-100-24 24V	1	102
21805100	CABLE FFC P=0.5mm 40V Type 2 (OPPOSITE CONTACTS) L=152mm (WE 687 740 152 002)	1	321
30114011	PHASE CABLE PLUG-FILTER	1	300
30114021	NEUTRAL CABLE PLUG-FILTER	1	301
30114191	GROUND CABLE L=150 O/F	2	302

30129530	HELIX HALF CABLE	1	318 - for NTC
30129670	MOT.DRV.DSP 8X AWG24 L=300mm CABLE	1	319
30135780	CABLED FUN MULTICOMP MC36288	1	105
30137930	OUT POWER&BUS BOARD	1	108
30137980	EXTERNAL CONNECTION BOARD (WITHOUT POWER SUPPLY)	1	106
30137990	POWER SUPPLY DC 4 POLE CABLE	1	304
30138680	CABLE FOR AC INPUT	1	303
30138820	MOVING ASSEMBLY INTERCONNECTION BOARD	1	109
30138991	CABL. FFC P=2.54mm 20V 55cm F-F INVERSE POLARIZATION UL	1	320
30139001	USB 2x L=250mm SHIELDED CABLE	1	305
30139010		1	306
30139021	POWER CPU CORTEX + PRINTER CABLE	1	307
30139031	SERIAL RS232 HOST VES 30 TOUCH L=400mm CABLE	1	308
30139042	SERIAL RS232 PRINTER & INTERNAL BARCODE CABLE	1	309
30139061	EXTENSION FOR WIRED COIL L = 650mm	1	310
30139071	MICROSWITCH COVER CABLE	1	311
30139081	MOLEX 6471 4P - MOLEX 6471 4P L=200mm CABLE	2	312
30139091	AMP MODU2 4V - AMP MODU2 4V L=200mm CABLE	2	313
30139100	DOUBLE READING SENSOR CABLE	1	314
30140901	MOLEX 5264 4P - MOLEX 5264 4P L=300mm CABLE	1	315
30141020	AMP MODU2 4P - AMP MODU2 4P MV L=550 CABLE	1	316
30141030	EXTENSION PHOTOCELL SCREEN L = 550 CABLE	1	317
30145580	CPU CORTEX M4 180MHz TFT + 3 STEP MOT v. CUBE30TOUCH	1	107
30146240	10.1 "DISPLAY CONNECTOR BOARD -INTERF.LVDS V.CUBE30TOUCH	1	110

NOTE: for the actually orderable part numbers refer to the official spare parts list

(a)

(b)

REGOLARE LO SLOT DELLA FOTOCELLULA CENTRATO RISPETTO ALLA LINGUETTA
SET THE SENSOR SLOT CENTERED WITH RESPECT TO THE PLASTIC TONGUE

REGOLARE LA POSIZIONE AL MONTAGGIO
ADJUST THE POSITION AT THE ASSEMBLY

VEDI NOTA (SEE NOTE)

201	202	203	401
002			
201	202	203	401
001			

P/N	Description	Q.ta	Reference
11141444	SCREW M4X20 UNI 5931 INOX A2	4	201
1114501Z	WASHER 4,3X9 UNI 6592 INOX A2	4	202
1114B37Z	WASHER A 4,3 UNI 1751 GROVER INOX	4	203
11400060	FLACONE LOCTITE FRENAFILETTI 243 (50 ml)	1	401
30140360	FFC P=1.27mm 10V 50cm F-F POLAR. INVERTED UL CABLE	1	300
30140370	EXTENSION CABLE L=350mm for DISCONNECTABLE PHOTOCELL	2	301
30223820	SWINGING MECHANICS SUPPORT ASS'Y	1	1
30223831	SWINGING MECHANICS ASS'Y	1	2

NOTE: for the actually orderable part numbers refer to the official spare parts list

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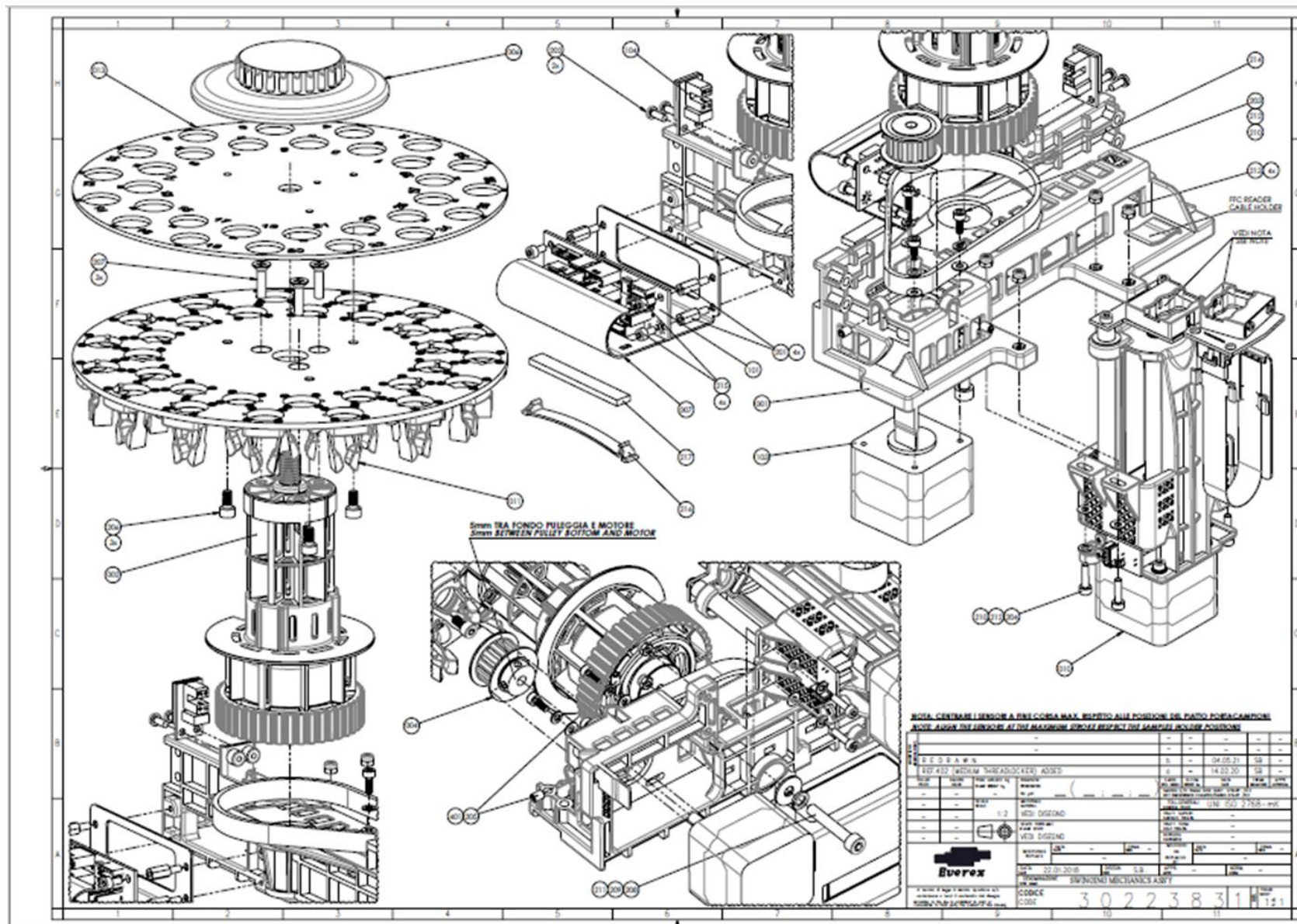


P/N	Description	Q.ta	Reference
101A4341	DRIVEN PULLEY SPACER	1	010
101A4360	SHORT PIVOT	1	002
101A4371	LONG PIVOT	1	003
101A4441	OVERTURNING BEARING FLANGE	1	006
101A6410	FULCRUM BUSHING	2	004
101A6450	MOTOR SPACER PLATE	1	007
101A6460	DRIVEN PULLEY T2.5-96 BORE Ø12	1	005
103A4502	INTERNAL BARCODE READER SUPPORT	1	008
103A6431	MECHANICS SUPPORT	1	001
11101152	SCREW M3X4 UNI 7687 ZN	3	201
11101937	SCREW M6X10 UNI 5925 Zn (STEI Estr.Cil.)	1	202
1110389H	SCREW PZAB 2,9X6,5 UNI 6955 ZN	1	203
1110456Z	PREVAILING TORQUE HEX.NUT M6 UNI 7474 ZN	1	204
1110F42Z	HEX. SPACER M-F M4X15	2	205
1110F56Z	STEAB SPACER 5021/3////06 - D7 d3,3 H3	2	206
1110H69Z	STEAB SPACER 5015/4/4//26 - D10 d4,3 H4	8	207
1110M03Z	CIRCLIP V-SEEGER FOR BORES JV24	2	208
1110M65Z	CYLINDRICAL SPACER 311321540050 (D6-d3.2-H15) BRASS Ni	4	209
11140882	SCREW M1.6X5 UNI 7687 INOX	2	210
11141156	SCREW M3X4 UNI 5923 INOX	2	211
11141183	SCREW M3X8 UNI 7688 INOX	6	212
11141204	SCREW M3X12 UNI 5931 INOX	2	213
11141274	SCREW M3X30 UNI 5931 INOX	4	214
11141384	SCREW M4X8 UNI 5931 INOX	2	215
11141423	SCREW M4X16 UNI 7688 INOX	2	216
11141424	SCREW M4X16 UNI 5931 INOX	2	217
1114459Z	HEX. LOCK NUT M3 UNI 7473 INOX	2	218
1114474L	HEX. NUT M4 UNI 5588 INOX	6	219
1114495Z	WASHER 2,2X5 UNI 6592 INOX	2	220
1114499Z	WASHER 3,2X7 UNI 6592 INOX A2	3	221
1114501Z	WASHER 4,3X9 UNI 6592 INOX A2	6	222
1114549Z	WASHER 6X18 INOX	1	223
1114B34Z	WASHER A 2,2 UNI 1751 GROVER INOX	2	224
1114B36Z	WASHER A 3,2 UNI 1751 GROVER INOX	4	225
1114B37Z	WASHER A 4,3 UNI 1751 GROVER INOX	2	226
11200120	SHIM RING 5x10x0,5	2	227

11400160	LOCTITE 222 50 ml	1	401
11504260	BALL BEARING 625 2Z	1	228
11505450	BALL BEARING 61802 (15x24x5)	2	229
11712410	BELT 6 T2,5/380 STEEL CORDS REINFORCED POLYURET.	1	230
20550A60	BARCODE READER LEUZE CR50M2/R2-S5	1	106
30119190	WIRED MOTOR 103-547-52500	1	103
30128420	SEPARABLE TEMP. SENSOR BOARD	1	105
30138690	DISCONNECTABLE PHOTOSENSOR BOARD TOP VERSION	1	102
30401170	SHIFT DRIVING PULLEY	1	009

NOTE: for the actually orderable part numbers refer to the official spare parts list

3.8.2 Swinging mechanics ass'y exploded view



P/N	Description	Q.ta	Reference
101A4451	DRIVING PULLEY 14XL d5	1	004
103A4590	CABLE GUIDE BRACKET	1	007
10605330	ADAPTER LOCKING KNOB	1	006
10605400	MOBILE MECHANICS MAIN SUPPORT	1	001
11000470	AERSTOP SE-40/A PROFILO ADES.8x3 NERO	0,06	217
1110027Z	HEX. SPACER M-F M3X10	4	201
1110546Z	WASHER M5X15 ZN	1	211
1110H85Z	SCREW TC ECO-SYN BN 82428 3x8 xTERMOPL.	2	202
11141184	SCREW M3X8 UNI 5931 INOX	4	203
11141204	SCREW M3X12 UNI 5931 INOX	4	204
11141376	SCREW M4X6 UNI 5923 INOX	2	205
11141384	SCREW M4X8 UNI 5931 INOX	3	206
11141423	SCREW M4X16 UNI 7688 INOX	3	207
11141714	VITE M5X30 UNI 5931 INOX	1	208
1114459Z	HEX. LOCK NUT M3 UNI 7473 INOX	4	213
1114499Z	WASHER 3,2X7 UNI 6592 INOX A2	7	210
1114B36Z	WASHER A 3,2 UNI 1751 GROVER INOX	7	212
1114B38Z	ROSETTA A 5,3 UNI 1751 GROVER INOX	1	209
11712480	BELT 114 XL 037 NEOPRENE/FIBERGLASS CORDS	1	214
20920630	CABLE HOLDER ESSENTA FCCS-5	1	216
30119190	WIRED MOTOR 103-547-52500	1	103
30133900	DISCONNECTABLE PHOTOSENSOR BOARD	1	104
30138820	MOVING ASSEMBLY INTERCONNECTION BOARD	1	101
30221402	DUAL CHANNEL READER ASS'Y	1	10
30221412	SAMPLES HOLDER 30x ASS'Y (GRAY HOLDERS)	1	011
30227260	42XL CENTRAL SHAFT ASS'Y	1	002
30404390	ENGRAVED POSITIONS MARKER DISC 30x	1	013

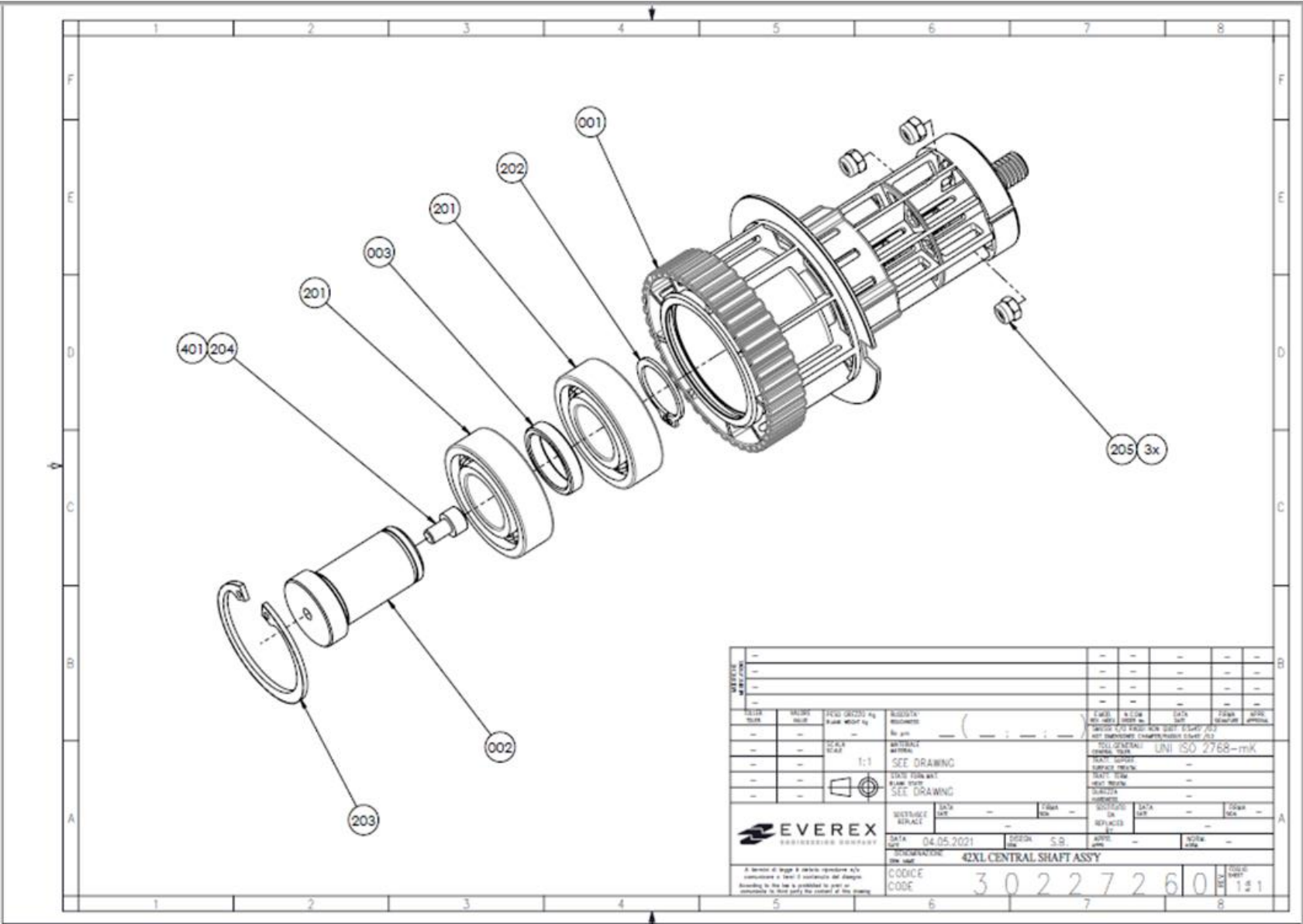
NOTE: for the actually orderable part numbers refer to the official spare parts list

[illegible]

P/N	Description	Q.ta	Reference
103A4411	SAMPLES HOLDER 30x	1	001
10608470	LINK SPRING (GRAY)	60	003
1110H76Z	SCREW TS ECO-SYN BN 82427 2,2x8 xTERMOPL	120	201
30405790	BOTTOM GRAY MACHINED HALF-LINK	30	002

NOTE: for the actually orderable part numbers refer to the official spare parts list

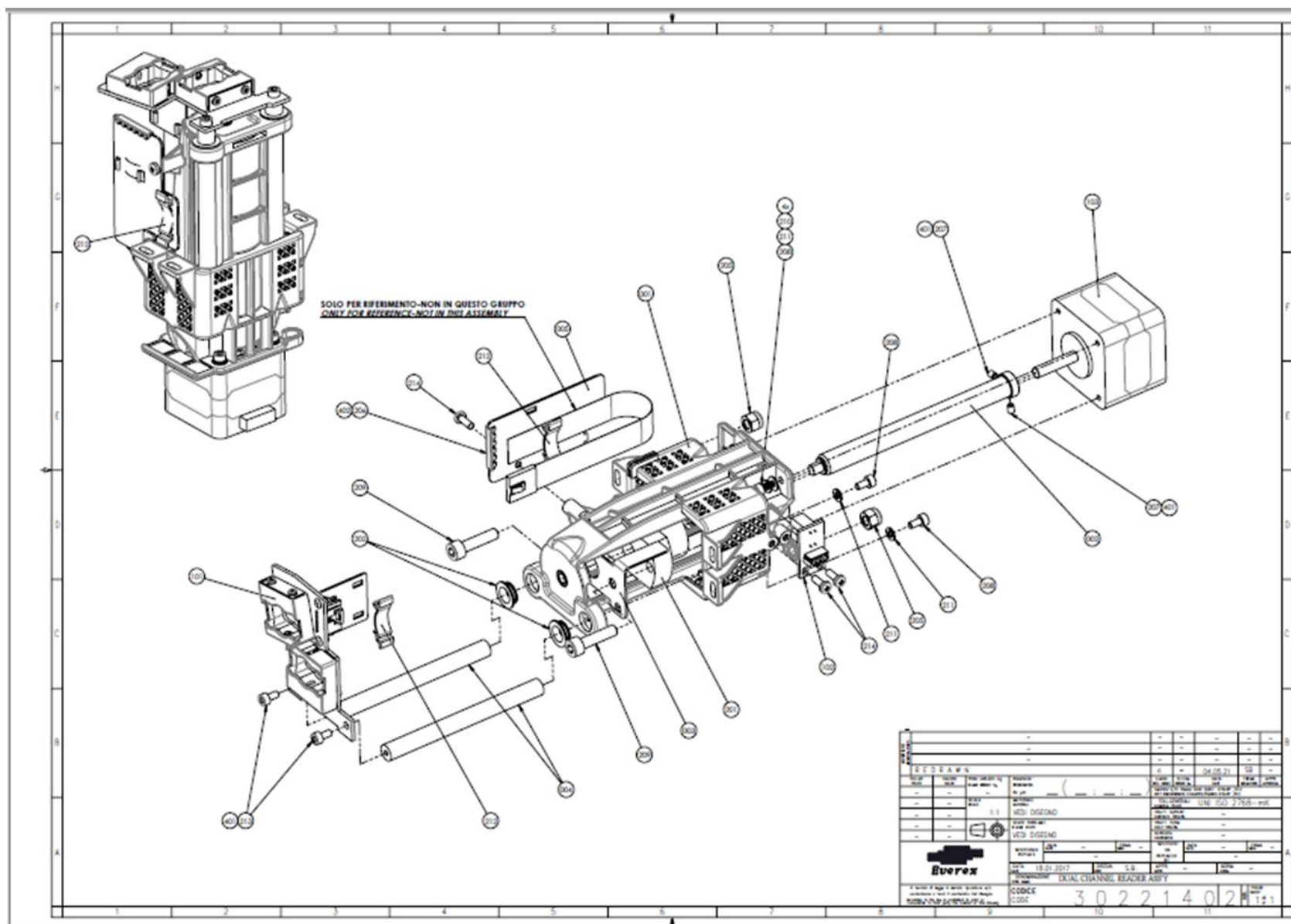
3.8.4 Central shaft ass.y



P/N	Description	Q.ty	Reference
10118761	DRIVEN PULLEY	1	001
10119011	SPACER	1	003
101C2580	HUB SHAFT	1	002
1110460Z	HEX. LOCK NUT M4 UNI 7473 ZN	3	205
1110596R	RETAINING RING E 20 UNI 7435	1	202
1110611T	ANELLO SEEGER I 42 UNI7437	1	203
11141614	SCREW M5X8 UNI 5931 INOX	1	204
11504370	BALL BEARING 6004-2Z	2	201
30405590	42XL MACHINED CENTRAL SHAFT	1	001

NOTE: for the actually orderable part numbers refer to the official spare parts list

3.9 Dual channel reading unit exploded view



P/N	Description	Q.ta	Reference
101A4541	NUT PLATE	1	003
101A4561	READER BOARD SPACER	2	004
103A4531	READER FFC SUPPORT	1	005
1110457Z	PREVAILING TORQUE HEX.NUT M5 UNI 7474 ZN	2	205
1110H85Z	SCREW TC ECO-SYN BN 82428 3x8 xTERMOPL.	3	214
1110L68Z	PROFILATE DX 1 CAF PLASTICA E.ZUBEHOR 35MT	23mm	206
1110M59Z	SCREW NUT IGUS DRYSPIN DST-JFRM-252525DS10x12	1	201
11141156	SCREW M3X4 UNI 5923 INOX	2	207
11141174	SCREW M3X6 UNI 5931 INOX	4	213
11141184	SCREW M3X8 UNI 5931 INOX	4	208
11141674	VITE M5X20 UNI 5931 INOX	2	209
1114499Z	WASHER 3,2X7 UNI 6592 INOX A2	4	210
1114B36Z	WASHER A 3,2 UNI 1751 GROVER INOX	6	211
11504570	CLIP BUSHING IGLIDUR MCM-08-02	2	202
20920600	FLAT CABLE CLAMP (SNAP-IN) RICHCO FCCS-1	3	212
30119190	WIRED MOTOR 103-547-52500	1	103
30133900	DISCONNECTABLE PHOTOSENSOR BOARD	1	102
30138951	DOUBLE SENSOR COMPLETE READING BOARD R.1	1	101
30230400	COMPLETE READER FRAME	1	001
30404332	READER SCREW 10x12	1	002

NOTE: for the actually orderable part numbers refer to the official spare parts list

Rev. 1.3 (07.2021)



P/N	Description	Q.ta	Reference
101A7181	DISPLAY SHELL	1	002
101B0060	DISPLAY BACK COVER	1	004
101B0070	INSULATOR 100x40	1	003
10803380	GROUND LABEL 13x13 UL	1	005
11000430	PIVILENE PROFILE BLACK ADHESIVE	0.14	006
1110F42Z	HEX. SPACER M-F M4X15	4	201
11141374	SCREW M4X6 UNI 5931 INOX	4	203
11141394	SCREW M4X10 UNI 5931 INOX	1	204
1114473L	HEX. NUT M3 UNI 5588 INOX	4	202
1114474L	HEX. NUT M4 UNI 5588 INOX	1	205
1114B62Z	SERRATED LOCK WASHER A 4,3 DIN 6798 INOX	3	206
12300760	ADHESIVE TAPE MACAL 8900 PRO BLACK MATT 20mm WIDTH	1	007
20510970	DISPLAY LCD TFT 10.1" 1024X600 TOUCH DLC01010EZG-T- 2 [DLC Display]	1	101
30114221	GROUND CABLE L=300 O/O	1	303
30136570	LCD WITH LEDs/EVX MOTHER BOARD CABLE	1	301
30140870	DISPLAY 10,1" INTERCONNECTION BOARD	1	102
30140890	CABLE FOR CONNECTION DISPLAY 10,1"	1	302
30141040	RGB LED STRIP WITH CABLE	1	103
30223800	CUBE 30 TOUCH COMPLETE FRONT PANEL	1	001*

NOTE: for the actually orderable part numbers refer to the official spare parts list

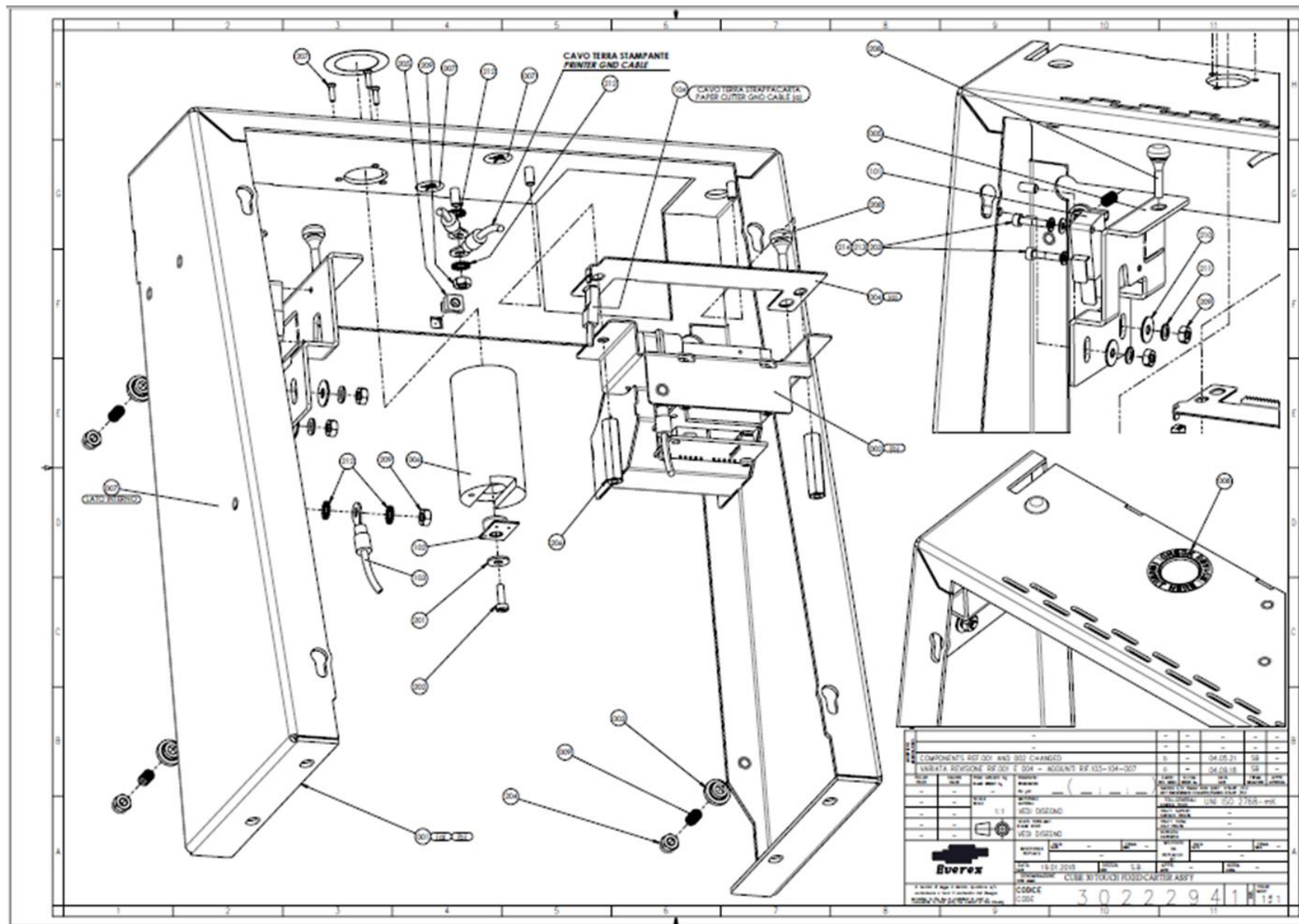
Rev. 1.3 (07.2021)



P/N	Description	Q.ta	Reference
103A4142	WELDED REMOVABLE CARTER	1	001
103A4172	WELDED LID	1	002
103A4231	LID SWITCH ACTUATOR	1	003
103A7301	LID STOPPER	1	004
103A7311	WELDED LID SUPPORT	1	005
10803380	GROUND LABEL 13x13 UL	1	006
10803880	"BIO HAZARD" LABEL 25x25 UL	1	007
1110H35Z	HEX SPACER FF10 H 4070 10 (M4X10) E.ZUB.	1	213
1110H96Z	SPACER Art. 341000 - 3460PA-02	4	201
1110L48Z	HEX. SPACER F-F M3x15	4	202
11141184	SCREW M3X8 UNI 5931 INOX	4	203
11141372	SCREW M4X6 UNI 7687 INOX	1	214
11141402	SCREW M4X12 UNI 7687 INOX	8	204
11141444	SCREW M4X20 UNI 5931 INOX A2	1	205
1114474L	HEX. NUT M4 UNI 5588 INOX	9	206
1114523Z	FLAT WASHER 3,2X9 UNI 6593 INOX	4	207
1114544Z	WASHER 4X12 INOX	6	208
1114B36Z	WASHER A 3,2 UNI 1751 GROVER INOX	4	209
1114B37Z	WASHER A 4,3 UNI 1751 GROVER INOX	7	210
1114B62Z	SERRATED LOCK WASHER A 4,3 DIN 6798 INOX	6	211
12500870	ADJUSTABLE TORQUE HINGE SOUTHCO E6-10-301-10 WHITE	2	212
30114211	GROUND CABLE L=200 O/O	1	301
10809960	CARTER ADHESIVE PROTECTION	2	008

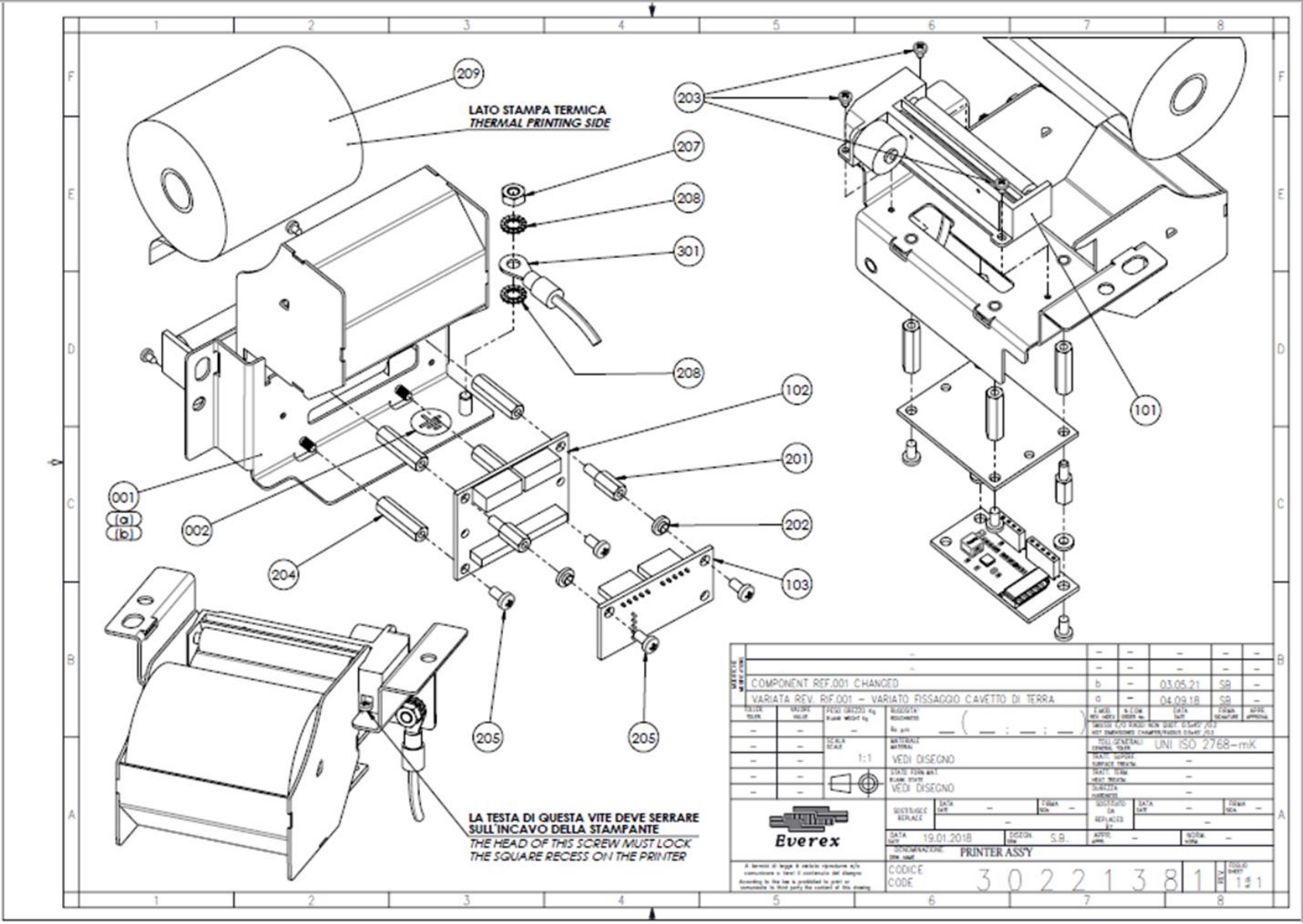
NOTE: for the actually orderable part numbers refer to the official spare parts list

3.12 Frontal fixed cover exploded view



P/N	Description	Q.ta	Reference
10127710	CUP FIXING BUSHING	4	003
101A4290	PAPER CUTTER	1	004
101A4321	LID SWITCH SUPPORT	1	005
101A4331	CHECK DEVICE LOADING PIT	1	006
103A4111	WELDED FIXED CARTER	1	001
10803380	GROUND LABEL 13x13 UL	1	007
10804200	CHECK DEVICE PIT LABEL	1	008
1110009Z	STEAB WASHER 5015/4/1/16	1	201
1110028Z	SCREW M3X10 STEAB 5074/10/25	1	202
1110392G	SCREW PZAB 2,9X16 UNI 6954 ZN	2	203
1110460Z	HEX. LOCK NUT M4 UNI 7473 ZN	4	204
1110H44Z	CAGE NUT M4 INOX (CLIP FOR THICK.2mm)	1	205
1110H46Z	HEX. SPACER FF10 H 4070 30 (M4x30) E.ZUB	2	206
1110H76Z	SCREW TS ECO-SYN BN 82427 2,2x8	3	207
1110M51Z	BUMPER ESSENTA BPM-2	2	208
1114474L	HEX. NUT M4 UNI 5588 INOX	3	209
1114544Z	WASHER 4X12 INOX	2	210
1114B37Z	WASHER A 4,3 UNI 1751 GROVER INOX	2	211
1114B62Z	SERRATED LOCK WASHER A 4,3 DIN 6798 INOX	2	212
11300570	MOLLA COMPR. De5 f=0,5 p=2 L=8	4	009
21200680	MICROSWITCH HONEYWELL V5C010BG1L	1	101
30139050	WIRED COIL	1	102
30221380	PRINTER ASS'Y	1	002

NOTE: for the actually orderable part numbers refer to the official spare parts list



P/N	Description	Q.ta	Reference
103A4261	WELDED PRINTER SUPPORT	1	001
10803380	GROUND LABEL 13x13 UL	1	002
1110027Z	HEX. SPACER M-F M3X10	2	201
1110082Z	INSULATOR TO220 STEAB 5160/12//118 ELMI	2	202
1110383G	VITE PZAB 2,2X6,5 UNI 6954 ZN - AF TCB IC	2	203
1110L42Z	DISTANZIALE ESAG. F-F M3x20	4	204
11141172	SCREW M3X6 UNI 7687 INOX	4	205
11141394	SCREW M4X10 UNI 5931 INOX	1	206
1114474L	HEX. NUT M4 UNI 5588 INOX	1	207
1114B62Z	SERRATED LOCK WASHER A 4,3 DIN 6798 INOX	2	208
12300000	THERMAL PAPER ROLLER H.MM L=57 D=50	1	209
21430340	THERMAL PRINTER PRT ART. PT486F-B	1	101
21430370	PRINTER CONTROLLER PRT PT486F (RS232)	1	102
30114201	GROUND CABLE L=120 O/O	1	301
30141160	SCHEDA REGOLATORE 8V PRINTER (Vin= 24V)	1	103

NOTE: for the actually orderable part numbers refer to the official spare parts list

3.13 Official spare part list

P/N Code	Descrizione	Description	REF.
30006202	STRUMENTO VES-MATIC CUBE 30 TOUCH	VES-MATIC CUBE 30 TOUCH INSTRUMENT	TAB. 3.6.
10127710	BOCCOLA FISSAGGIO VASCHETTA	CUP FIXING BUSHING	TAB. 3.12
101A4321	SUPPORTO SWITCH SPORTELLLO	LID SWITCH SUPPORT	TAB. 3.12
103A4111	CARTER FISSO SALDATO	WELDED FIXED CARTER	TAB. 3.12
11300570	MOLLA COMPR. De5 f=0,5 p=2 L=8	MOLLA COMPR. De5 f=0,5 p=2 L=8	TAB. 3.12
21200680	MICROSWITCH HONEYWELL V15T16-CZ200A05 UL	MICROSWITCH HONEYWELL V15T16-CZ200A05 UL	TAB. 3.12
30139050	WIRED COIL SCONNETTIBILE	WIRED COIL	TAB. 3.12
30221361	ASSIEME BASAMENTO	BASEMENT ASS'Y	TAB. 3.7.
10139151	ISOLANTE PER ALIMENTATORE 100W UL	POWER SUPPLY INSULATOR (100W)	TAB. 3.7.
103A4083	BASAMENTO SALDATO	WELDED BASEMENT	TAB. 3.7.
103A4221	FONDO BASAMENTO	BASEMENT BOTTOM	TAB. 3.7.
10807850	ETICHETTA CONNESSIONI ESTERNE VMC30T	VMC30T EXTERNAL CONNECTIONS LABEL	TAB. 3.7.
21200540	SWITCH BIP. LIGHT C. R5-1-5-K-K-F-T-O-F UL	SWITCH BIP. LIGHT C. R5-1-5-K-K-F-T-O-F	TAB. 3.7.
21201080	FILTRO RETE SCHAFFNER FN9260-10-06 UL	FILTER FOR P.SUPPLY SCHAFFNER FN9260-10-06 UL	TAB. 3.7.
21445000	ALIM. MEAN-WELL LPS-100-24 24V 4.2A UL	SW.POWER SUPPLY MEAN-WELL LPS-100-24 24V	TAB. 3.7.
30114011	CAVO FASE SPINA/FILTRO UL	PHASE CABLE PLUG-FILTER	TAB. 3.7.
30114021	CABL. NEUTRO SPINA/FILTRO UL	NEUTRAL CABLE PLUG-FILTER	TAB. 3.7.
30114191	CABL.TERRA L=150 O/F UL	GROUND CABLE L=150 O/F	TAB. 3.7.
30129530	CABL. MOLEX 5264 2P DRITTO L=600	CABLE 2 POLE STRAIGHT	TAB. 3.7.
30129670	CABL. PER MOT.DRV.DSP 8X AWG24 L=300mm	CABL. PER MOT.DRV.DSP 8X AWG24 L=300mm	TAB. 3.7.
30135780	VENTOLA MULTICOMP MC36288 CABLATA	FAN WIRED	TAB. 3.7.
30136901	SCHEDA CPU CORTEX M4 180MHz TFT + 3 STEP MOT	CPU CORTEX M4 180 MHz TFT + 3 STEP MOT BOARD	TAB. 3.7.
30137930	SCHEDA USCITA ALIMENTAZIONI&BUS	POWER SUPPLY CONNECTION BOARD	TAB. 3.7.
30137980	SCHEDA RACCORDO CONNESSIONI ESTERNE NO ALIM.	CONNECTION BOARD	TAB. 3.7.
30137990	CABL. ALIM. DC OUT+24V MW-LPP 4POLI	POWER SUPPLY DC 4 POLE CABLE	TAB. 3.7.
30138680	CABL. BOX ALIM. INGRESSO 220 VAC UL x LPS	POWER SUPPLY AC INLET CABLE	TAB. 3.7.
30138820	SCHEDA RACCORDO MOVIMENTAZIONE VMC30 NEW	CONNECTION FOR MOVABLE CABLE BOARD	TAB. 3.7.
30138991	CABL. FFC P=2.54mm 20V 55cm F-F POLAR. INVERSA UL	CABLE FFC P=2.54mm 20V 55cm F-F UL	TAB. 3.7.
30139001	CABL. USB 2x L=220mm SCHERMATO	CABLE USB 2x L=220mm WITH SHIELD	TAB. 3.7.
30139010	CABL. ALIM.LOGICA&SERIALE RS232 CPU CORTEX	CABLE LOGIC PWR SUPPLY & SERIAL RS232 x CPU CORTEX	TAB. 3.7.

30139021	CABL. ALIM.POTENZA CPU CORTEX + PRINTER	CABLE POWER SUPPLY FOR CPU CORTEX AND PRINTER	TAB. 3.7.
30139031	CABL. SERIALE RS232 HOST VES 30 TOUCH L=400mm	CABLE FOR SERIAL RS232 HOST VES 30 TOUCH L=400mm	TAB. 3.7.
30139042	CABL. SERIALE RS232 PRINTER & INTERNAL BARCODE	CABLE FOR SERIAL RS232 PRINTER AND INTERNAL BARCODE	TAB. 3.7.
30139061	PROLUNGA PER WIRED COIL L=650mm	CABLE FOR WIRED COIL L=650mm	TAB. 3.7.
30139071	CABL. MICROSWITCH COVER	CABLE MICROSWITCH COVER	TAB. 3.7.
30139081	CABL. MOLEX 6471 4P - MOLEX 6471 4P L=200mm	CABLE MOLEX 6471 4P - MOLEX 6471 4P L=200mm	TAB. 3.7.
30139091	CABL. AMP MODU2 4V - AMP MODU2 4V L=200mm	CABLE AMP MODU2 4V - AMP MODU2 4V L=200mm	TAB. 3.7.
30139100	CABL. DOPPIO SENSORE LETTURA	CABLE FOR DOUBLE READING SENSOR	TAB. 3.7.
30140380	SCHEDA RACCORDO DISPLAY 10.1"-INTERF.LVDS	CONNECTION BOARD FOR DISPLAY 10.1" LVDS INTERFACE	TAB. 3.7.
30140901	CABL. MOLEX 5264 4P - MOLEX 5264 4P L=300mm	CABLE MOLEX 5264 4P - MOLEX 5264 4P L=300mm	TAB. 3.7.
30141020	CABL. AMP MODU2 4P - AMP MODU2 4P MV L=550	CABLE AMP MODU2 4P - AMP MODU2 4P MV L=550	TAB. 3.7.
30141030	CABL. PROLUNGA FOTOCELLULA SCONNETT. L=550	CABLE FOR PHOTOCOUPLER SENSOR L=550	TAB. 3.7.
30221380	ASSIEME STAMPANTE	PRINTER ASS'Y	TAB. 3.12.1
101A4290	STRAPPACARTA	PAPER CUTTER	TAB. 3.12
103A4261	SUPPORTO STAMPANTE SALDATO	WELDED PRINTER SUPPORT	TAB. 3.12.1
12300000	ROTOLO CARTA TERMICO H.MM L=57 D=50	THERMAL PAPER ROLLER H.MM L=57 D=50	TAB. 3.12
21430340	STAMPANTE TERMICA PRT ART. PT486F-B	THERMAL PRINTER PRT ART. PT486F-B	TAB. 3.12
21430370	CONTROLLER STAMPANTE PRT PT486F (RS232)	PRINTER CONTROLLER PRT PT486F (RS232)	TAB. 3.12
30114201	CABL.TERRA L=120 O/O UL	GROUND CABLE L=120 O/O	TAB. 3.12
30141160	SCHEDA REGOLATORE 8V PRINTER (Vin= 24V)	PRINTER POWER SUPPLY BOARD	TAB. 3.12
30221391	ASSIEME MECCANICA INTERNA	INTERNAL MECHANICS ASS'Y	TAB. 3.8.
101A4341	DISTANZIALE PULEGGIA CONDOTTA	DRIVEN PULLEY SPACER	TAB. 3.8.1
101A4360	PERNO CORTO	SHORT PIVOT	TAB. 3.8.1
101A4371	PERNO LUNGO	LONG PIVOT	TAB. 3.8.1
101A4382	ANGOLARE MECCANICA MOBILE	MOBILE MECHANICS ANGULAR	TAB. 3.8.2

101A4391	MOZZO PIATTO	SAMPLES HOLDER HUB	TAB. 3.8.2
101A4400	DISCO SENSORE HOME	HOME SENSOR DISC	TAB. 3.8.2
101A4441	FLANGIA CUSCINETTO RIBALTAMENTO	OVERTURNING BEARING FLANGE	TAB. 3.8.1
101A4451	PULEGGIA MOTRICE 14XL d5	DRIVING PULLEY 14XL d5	TAB. 3.8.2
101A4511	SUPPORTO MOTORE ROTAZIONE	ROTATION MOTOR SUPPORT	TAB. 3.8.2
101A6410	BUSSOLA FULCRO	FULCRUM BUSHING	TAB. 3.8.1
101A6450	PIASTRA DISTANZIALE MOTORE	MOTOR SPACER PLATE	TAB. 3.8.1
101A6460	PULEGGIA CONDOTTA T2.5-96 FORO Ø12	DRIVEN PULLEY T2.5-96 BORE Ø12	TAB. 3.8.1
101A7190	POMELLO SUPERIORE	TOP KNOB	TAB. 3.8.2
103A4502	SUPPORTO BARCODE INTERNO	INTERNAL BARCODE READER SUPPORT	TAB. 3.8.1
103A4590	STAFFA GUIDACAVO	CABLE GUIDE BRACKET	TAB. 3.8.2
103A4601	SUPPORTO SENSORE HOME	HOME SENSOR SUPPORT	TAB. 3.8.2
103A4611	ATTUATORE SENSORE RIBALTAMENTO	OVERTURNING SENSOR ACTUATOR	TAB. 3.8.2
103A6431	SUPPORTO MECCANICA	MECHANICS SUPPORT	TAB. 3.8.1
11504260	CUSCINETTO 625 2Z	BALL BEARING 625 2Z	TAB. 3.8.1
11505450	CUSCINETTO 61802 (15x24x5)	BALL BEARING 61802 (15x24x5)	TAB. 3.8.1
11712410	CINGHIA 6 T2,5/380 POLIURET. RINFORZ. ACCIAIO	BELT 6 T2,5/380 STEEL CORDS REINFORCED POLYURET.	TAB. 3.8.1
11712480	CINGHIA IN GOMMA 114 XL 037 - TREFOLI FIBRA VETRO	BELT 114 XL 037 NEOPRENE/FIBERGLASS CORDS	TAB. 3.8.2
20550A60	BARCODE READER LEUZE CR50M2/R2-S5	BARCODE READER LEUZE CR50M2/R2-S5	TAB. 3.8.1
30119190	MOTORE 103-547-52500 CABLATO	WIRED MOTOR 103-547-52500	TAB. 3.8.1
30128420	SCHEDINO SENSORE TEMPERATURA SCONNETTIBILE	SEPARABLE TEMP. SENSOR BOARD	TAB. 3.8.1
30133900	SCHEDA FOTOCELLULA SCONNETTIBILE	DISCONNECTABLE PHOTOSENSOR BOARD	TAB. 3.8.2
30138690	SCHEDA FOTOCELLULA SCONNETTIBILE V.TOP	DISCONNECTABLE PHOTOSENSOR BOARD V.TOP	TAB. 3.8.1
30138820	SCHEDA RACCORDO MOVIMENTAZIONE VMC30 NEW	CONNECTION FOR MOVABLE CABLE BOARD	TAB. 3.8.2

30140360	CABL. FFC P=1.27mm 10V 50cm F-F POLAR. INVERSA UL	CABLE FFC P=1.27mm 10V 50cm F-F UL	TAB. 3.8.
30140370	CABL. PROLUNGA FOTOCELLULA SCONNET. L=350mm	CABLE FOR PHOTOSENSOR L=350 mm	TAB. 3.8.
30401170	PULEGGIA MOTRICE AVANZAMENTO	SHIFT DRIVING PULLEY	TAB. 3.8.
30221401	ASSIEME LETTORE DOPPIO CANALE	DUAL CHANNEL READER ASS'Y	TAB. 3.9.
101A4541	PIASTRINA PER CHIOCCIOLA	NUT PLATE	TAB. 3.9
101A4561	DISTANZIALE SCHEDA LETTORE	READER BOARD SPACER	TAB. 3.9
30223810	CORPO LETTORE CON BOCCOLA	READER MAIN FRAME WITH BUSHING	TAB. 3.9
101A5181	PIASTRINA GUIDA LETTORE	READER GUIDE PLATE	TAB. 3.9
101A5970	TASSELLO NEOPRENE ADESIVO	NEOPRENE ADHESIVE SCRAP	TAB. 3.9
103A4521	SUPPORTO LETTORE DOPPIO CANALE	DUAL CHANNEL READER SUPPORT	TAB. 3.9
103A4530	SUPPORTO CAVO FFC LETTORE	READER FFC SUPPORT	TAB. 3.9
11504570	BOCCOLA A CLIP IGLIDUR PER LAM.MCM-08-02	CLIP BUSHING IGLIDUR MCM-08-02	TAB. 3.9
30119190	MOTORE 103-547-52500 CABLATO	WIRED MOTOR 103-547-52500	TAB. 3.9
30133900	SCHEDA FOTOCELLULA SCONNETTIBILE	DISCONNECTABLE PHOTOSENSOR BOARD	TAB. 3.9
30138950	SCHEDA DOPPIO SENSORE LETTURA COMPLETO	DOUBLE READING SENSOR BOARD	TAB. 3.9
30221410	ASSIEME PIATTO PORTACAMPIONI 30x	SAMPLES HOLDER 30x ASS'Y	TAB. 3.8.3
103A4411	SAMPLES HOLDER 30x	DISCO PORTACAMPIONI 30x	TAB. 3.8.3
10602101	MOLLA RITEGNO CUVETTA (BLU)	LINK SPRING (BLU)	TAB. 3.8.3
30403310	SEMIMAGLIA INFERIORE BLU LAVORATA	BOTTOM HALF-LINK BLUE MACHINED	TAB. 3.8.3
30404390	DISCO SEGNAPOSIZIONI 30x INCISO	ENGRAVED POSITIONS MARKER DISC 30x	TAB. 3.8.2
30222100	ASSIEME PULEGGIA CONDOTTA	DRIVEN PULLEY ASS'Y	TAB. 3.8.4
10118761	PULEGGIA CONDOTTA	DRIVEN PULLEY	TAB. 3.8.4
10118781	PERNO PIATTO VES 30	PERNO PIATTO VES 30	TAB. 3.8.4
10119011	DISTANZIALE	SPACER	TAB. 3.8.4
1110596R	ANELLO SEEGER E 20 UNI 7435	RETAINING RING E 20 UNI 7435	TAB. 3.8.4
1110611T	ANELLO SEEGER I 42 UNI7437	ANELLO SEEGER I 42 UNI7437	TAB. 3.8.4
11504370	CUSCINETTO 6004-2Z	BALL BEARING 6004-2Z	TAB. 3.8.4
30222091	ASSIEME FRONTALE VES-MATIC CUBE 30 TOUCH	VES-MATIC CUBE 30 TOUCH FRONT PANEL ASS'Y	TAB. 3.10
101A7181	GUSCIO DISPLAY	DISPLAY SHELL	TAB. 3.10
101B0060	COVER POSTERIORE DISPLAY	DISPLAY BACK COVER	TAB. 3.10

101B0070	ISOLANTE 100x40	INSULATOR 100x40	TAB. 3.10
10807411	GRAFICA SAGOMATA VES-MATIC CUBE 30 TOUCH	VES-MATIC CUBE 30 TOUCH SCREENPRINTED PANEL	TAB. 3.10*
10808090	PANNELLO FRONTALE STAMPATO E LAMINATO	PRINTED AND LAMINATE FRONT PANEL	TAB. 3.10*
20510970	DISP. LCD TFT 10.1" 1024X600 TOUCH DLC01010EZG-T-2	DISP. LCD TFT 10.1" 1024X600 TOUCH DLC01010EZG-T-2	TAB. 3.10
30114221	CABL.TERRA L=300 O/O UL	GROUND CABLE L=300 O/O	TAB. 3.10
30136570	LCD WITH LEDS/EVX MOTHER BOARD CABLE	LCD WITH LEDS/EVX MOTHER BOARD CABLE	TAB. 3.10
30140870	SCHEDA RACCORDO DLC01010EZG-T-2 FULL SMD	DISPLAY 10,1" INTERCONNECTION BOARD	TAB. 3.10
30140890	CABL. RACCORDO DISPLAY DLC01010EZG-T-2	CABLE FOR CONNECTION DISPLAY 10,1"	TAB. 3.10
30141040	STRISCIA LED RGB CABLATA L=1000	RGB LED STRIP WITH CABLE	TAB. 3.10
30222400	ASSIEME CARTER CUBE 30 TOUCH	CUBE 30 TOUCH CARTER ASS'Y	TAB. 3.11
103A4142	CARTER MOBILE SALDATO	WELDED REMOVABLE CARTER	TAB. 3.11
103A4171	SPORTELLLO SALDATO	WELDED LID	TAB. 3.11
103A4231	ATTUATORE SWITCH SPORTELLLO	LID SWITCH ACTUATOR	TAB. 3.11
103A7301	BATTUTA SPORTELLLO	LID STOPPER	TAB. 3.11
103A7311	SUPPORTO SPORTELLLO SALDATO	WELDED LID SUPPORT	TAB. 3.11
12500870	CERNIERA A FRIZIONE REG. SOUTHCO E6-10-301-10 B	ADJUST. TORQUE HINGE SOUTHCO E6-10- 301-10 WHITE	TAB. 3.11
30114211	CABL. TERRA L=200 O/O	CABL. TERRA L=200 O/O	TAB. 3.11
30222480	ASSIEME BLOCCO PER SPEDIZIONE	SHIPPING LOCK ASS'Y	TAB. 3.6
30508340	<i>KIT ACCESSORI CUBE 30 TOUCH</i>	<i>CUBE 30 TOUCH ACCESSORIES KIT</i>	<i>TAB. 3.6</i>
101A6470	DISCO STOP RUBBER	RUBBER STOP DISC	
21890040	CAVO ALIM. 3x0.75mmq (AWG18) L=2m SCHUKO 90° - C13	220Vac CABLE 3x0.75 L=2m SCHUKO 90° - C1	
21890360	CAVO USB 180 LC HiSpeed 2.0 GREY 68712 / CAVO USB 1.1 A-B M/M 1.5mt	USB CABLE 1.1 A-B M/M 1,5 Mt	
21890370	CAVO ALIM. SVT=3x0.75mmq (AWG18) SPINA USA/PRESA VDE 2MT UL	220Vac CABLE SVT PLUG USA/SOCKET VDE 2MT	
10809960	PROTEZIONI ADESIVE DEL CARTER	CARTER ADHESIVE PROTECTION	TAB. 3.11

4 EXTERNAL CONNECTIONS



The cables used for the external connections must not exceed 3 meters in length.

4.1 General specifications for connecting to a barcode reader

Before connecting the external barcode reader it is recommended checking that:

1. This is fitted with a cable with a DB9 female connector in DTE set-up with a 5Vdc 9-pin power supply (refer to the instruction manual of the barcode reader),
2. The signals on the DB9 female connector are compatible with the connector installed on the rear of the instrument to which it is connected:

DB9 Male "EXTERNAL BARCODE" on system rear	
PIN	SIGNAL
2	[Tx] Data transmission towards the reader (not used)
3	[RX] Date received from the reader
5	GND
9	+ 5 V

TECHNICAL INFORMATION:

- The electrical levels of the signals are of the standard RS232 type.
- The communication is one-way from the barcode reader towards the machine.
- The speed of the transmission is 9600 bit/s; the data format is of the type with 8 data bits, 1 stop bit and no parity bit.
- The communication protocol is of the ASCII type; the barcode read must be terminated with the Carriage Return (0x0d).

Connect the instrument to the external barcode.

Use the Barcode Model Z-3080 [Zebex] (or equivalent) with a connection cable type CAB 50607-R9 [Zebex] (or equivalent).

4.2 Connection to a host computer

Foreword: Hardware data

The Communication between the CUBE 30 touch and an external PC may be done in two possible ways:

1. Using a **USB Connection**:

Connect a standard A-B USB cable between the computer's USB port (type-A rectangular connector) and the CUBE 30 touch's USB port (type-B rectangular connector). The driver (STM32 SW; download from www.diesse.it) for MS Windows will need to be installed to establish communication with the CUBE 30 touch through a virtual COM port on USB.

HOST BY USBD   On the Cube 30 touch, in the Service menu, the "HOST BY USBD" parameter must be set to ON.

2. Using a serial **RS232 COM** port on the PC.

Connect a straight standard serial cable between the PC's RS232 COM port and the instrument's RS232 (9-pin) serial connector.

HOST BY USBD   On the Cube 30 touch, in the Service menu, the "HOST BY USBD" parameter must be set to OFF.

The electrical levels of the signals are of the standard RS232C type.

- The default transmission speed is 9600 bit/s
- the data format is the 8-bit
- 1 stop bit and no parity bits
- The DB9 Male "RS232C" connector reflects the following pin-out:

PIN	SIGNAL
2	Rx of data from Host
3	Tx of data towards Host
5	GND

Foreword: Hexadecimal ASCII (HEX-ASCII) representation

In the protocol described below a great deal of the parameters and data are represented in Hexadecimal ASCII (HEX-ASCII) format, in other words: a byte with a value of 0x7A is represented by two ASCII characters: '7' (0x37) and 'A' (0x41), the first represents the most significant nibble and the second, the least significant.

Examples:

Original Byte	Representation HEX-ASCII	
Hexadecimal value	H characters	L characters
0x45	'4' (0x34)	'5' (0x35)
0xC8	'C' (0x43)	'8' (0x38)
0x6F	'6' (0x36)	'F' (0x46)
0x10	'1' (0x31)	'0' (0x30)

As can be seen, this type of representation means that two ASCII characters are necessary for representing the value of one byte.

General remarks: Delay in replying

To allow the machine time to activate the reception mode it is necessary to enter a delay of 1 second on the reply.

Message for sending Results: Command 0x51

This message is sent by the CUBE30TOUCH to the host computer. The message contains the results of the analyses carried out on one or more tubes. The host computer must reply to this message only with a message of the ACK or NACK kind to notify the successful receipt of the results or the presence of errors message.

N.B.: the samples with the attribute "unknown code" that have been analysed by the instrument are not automatically sent at the end of the analysis process, instead these can only be sent manually by the operator by pressing the "Send to Host" key on the Database of Pending Samples menu.

Command: CUBE 30 touch sends the following frame:

STX (0x3E)	H-BLK (0x30)	L-BLK (0x30)	H-LEN	L-LEN	H-ADD (0x30)	L-ADD (0x31)	H-COM (0x35)	L-COM (0x31)	Data-1	...	Data-n	ETX (0x0D)	H-CHK	L-CHK
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The hexadecimal values indicated in brackets are constant values for this message. The fields in bold print are the variable ones and are described below:

H-LEN / L-LEN:

Length of the data field, from Data-1 to Data-n inclusive, represented in HEX-ASCII. Maximum value 'F' (0x46) / 'F' (0x46). This is the effective number of bytes contained in the data field. The maximum number of bytes contained in the 'Data' field is in fact 255.

Data-1 .. Data-n:

Data field. The data field for the message code 0x51 consists of the following:

H-PRO / L-PRO (2 bytes HEX-ASCII)	Record Tube-1		Record Tube-n
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H-PRO / L-PRO: Number of Record Tube contained in the message, represented in HEX-ASCII.

The Record Tube number contained in the data field is limited by the fact that the data field itself is able to contain up to a maximum of 255 bytes, in any case the Record Tubes are never cut off.

Record Tubes:

BarCode (ASCII string max 15 characters)	Terminator of the string Barcode (0x10)	DATA ANALYSES ASCII string 6 characters	TIME ANALYSES ASCII string 4 characters	VES ASCII string 4 characters	H-FLAGS	L-FLAGS	RACK ID ASCII string 4 characters	POSITION ASCII string 2 characters
--	---	---	---	---	----------------	----------------	---	--

BARCODE: ASCII string with variable lengths, maximum of 15 characters allowed. This is the barcode just as it is read by the CUBE 30 touch Barcode Reader.

TERMINATOR: the string of barcodes terminates with the 0x10 byte. This is because the length of this string is variable.

DATA ANALYSES : string of 6 characters without terminator, "DDMMYY" where:

- "DD" = day of the month, from "01" to "31" ASCII.
- "MM" = Month of the year, from "01" to "12" ASCII.
- "YY" = Year of the century, from "00" to "99" ASCII.

TIME ANALYSES : string of 4 characters without terminator, “hhmm” where:

- “hh” = hour of the day, from “00” to “23” ASCII.
- “mm” = Minutes, from “00” to “59” ASCII.

VES: Value of the VES measured, ASCII string without terminator: from “ 0” (3 spaces + ‘0’) transmitted in the case of an error, to “ 140” (1 space + “140”). If the result is greater of 140 the string will be “>140”.

EXAMPLES, see following table:

VES value	String sent	Bytes of the String
1	“ 1”	0x20, 0x20, 0x20, 0x31
100	“ 100”	0x20, 0x31, 0x30, 0x30
>140	“>140”	0x3E, 0x31, 0x34, 0x30

H-FLAGS / L-FLAGS: Bitmap with 8-bit of the sample errors, represented in HEX-ASCII.

The following table illustrates the errors:

Bit	Error	Description
0	Sample High	Blood column too high
1	Sample Low	Blood column too low
2	Sample Absent	Tube Empty
3	Reading Error	Generic reading error
4	QC PASS	Reserved for samples with control blood
5	QC FAIL	Reserved for samples with control blood
6-7	-	Reserved

EXAMPLES:

- In the case of a “Sample High” error the Bit 0 (least significant) will be set to one and all the others to zero, therefore the byte of the Flags will have a 0x01 hexadecimal value and its HEX-ASCII representation will be 0x30 / 0x31.
- In the case of a “Sample Absent” error the Bit 2 will be set to one and all the others to zero, therefore the byte of the Flags will have a 0x04 hexadecimal value and its HEX- ASCII representation will be 0x30 / 0x34.

Managing Errors in RESULTS:

If a Test Tube Record is sent with a VES value equal to 0 and an Error Flag enabled (Bit 3 set to 1), the result (VES=0) must be interpreted by the Host as a 'Reading Error of the sample'.

RACK ID: Not Used, always "0000"(string of 4 characters without terminator)

POSITION: string of 2 characters without terminator, identifies the position occupied by the sample into the instrument ("01"....."04").

H-CHK / L-CHK:

Checksum of the message, represented in HEX-ASCII. The Checksum is calculated by carrying out the OR- exclusive of all the bytes sent from STX to ETX inclusive. The resulting hexadecimal value is then converted into HEX-ASCII and the two characters that represent it are sent.

ATTENTION: for debugging purposes it is possible to disable the checksum control, replacing the H-COM bytes with the value of 0x44 instead of 0x35. In this case the two bytes of the checksum are still sent but their value will be insignificant. The Host computer must also manage any possible cases in which the checksum is disabled.

Reply from the Host computer (optional)

On receiving the message, the Host computer may send an ACK reply to acknowledge correct receipt and interpretation of the message, meaning that all the fields have the correct values and the checksum is correct; or a NACK reply to indicate that the message contains one of more errors: inexact checksum, incorrect length of the data field, etc..

ACK Message

ACK (0x06)	H-ADD (0x30)	L- ADD (0x31)	ETX (0x0D)
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Timeout on ACK Message: 1 Sec.

NACK Message

NACK (0x15)	H-ADD (0x30)	L-ADD (0x31)	H-ERR	L-ERR	ETX (0x0D)
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where: H-ERR / L-ERR are the HEX-ASCII representation of the error code defined according to the following table:

Error code	H-ERR Value	L-ERR Value	Meaning
0x00	0x30	0x30	General Error
0x04	0x30	0x34	Checksum Error
0x05	0x30	0x35	Error field value H-LEN / L-LEN
0x06	0x30	0x36	Data field Length Error

Message for sending QC (Quality Control) sample data: Command 0x52

This message is sent by the CUBE 30 touch towards the host computer. The message contains the results of the analyses performed on one or more samples. The host computer must only reply to this message with an ACK or NACK type message to notify the successful receipt of the results or the presence of errors in the message.

Command: CUBE 30 touch sends the following frame:

STX (0x3E)	H-BLK (0x30)	L-BLK (0x30)	H-LEN	L-LEN	H-ADD (0x30)	L-ADD (0x31)	H-COM (0x35)	L-COM (0x32)	Data-1	...	Data-n	ETX (0x0D)	H-CHK	L-CHK
---------------	-----------------	-----------------	--------------	--------------	-----------------	-----------------	-----------------	-----------------	---------------	-----	---------------	---------------	--------------	--------------

The hexadecimal values indicated in brackets are constant values for this message. The fields in bold print are the variable ones and are described below:

H-LEN/L-LEN:

Length of the data field, from Data-1 to Data-n inclusive, represented in HEX-ASCII. Maximum Value 'F' (0x46) / 'F' (0x46). This is the effective number of bytes contained in the data field. In fact, the maximum number of bytes contained in the DATA field is 255.

Data-1.. Data-n:

Data field. The Data field for the message code 0x52 consists of the following:

QC data	QC Record Sample
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QC data

Batch No. (ASCII string 6 characters)	EXPIRY DATE ASCII string 6 characters	H-VALMIN	L-VALMIN	H-VALMAX	L-VALMAX
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BATCH No.: ASCII string of 6 characters. Identifies the production batch of the control blood

EXPIRY DATE: string of 6 characters without terminator, "DDMMYY" where:

- "DD" = day of the month, from "01" to "31" ASCII.
- "MM" = Month of the year, from "01" to "12" ASCII.
- "YY" = Year of the century, from "00" to "99" ASCII.

H-VALMIN / L-VALMIN: Value lower than the acceptability range for the control blood, represented in HEX-ASCII.

H-VALMAX / L-VALMAX: Value higher than the acceptability range for the control blood, represented in HEX-ASCII.

QC Record Sample:

BarCode (ASCII string max 15 characters)	Terminator of the Barcode string (0x10)	DATA ANALYSES ASCII string 6 characters	TIME ANALYSES String ASCII 4 characters	VES ASCII string 4 characters	H-FLAGS	L-FLAGS	RACK ID String ASCII 4 characters	POSITION String ASCII 2 characters
---	---	---	--	--	---------	---------	---	--

BARCODE: ASCII string with variable length, maximum of 15 characters allowed. This is the barcode just as it is read by the CUBE 30 touch Barcode Reader.

TERMINATOR: the string of barcodes terminates with the 0x10 byte. This is because the length of this string is variable.

DATA ANALYSES: string of 6 characters without terminator, "DDMMYY" where:

- "DD" = day of the month, from "01" to "31" ASCII.
- "MM" = Month of the year, from "01" to "12" ASCII.
- "YY" = Year of the century, from "00" to "99" ASCII.

TIME ANALYSES: string of 4 characters without terminator, "hhmm" where:

- "hh" = hour of the day, from "00" to "23" ASCII.
- "mm" = Minutes, from "00" to "59" ASCII.

VES: Value of the VES measured on the QC sample, ASCII string without terminator:

from “ 0” (3 spaces + ‘0’) transmitted in the case of an error, to “ 140” (1 space + “140”). If the result is higher than 140 the string will be “>140”

EXAMPLES, see following table:

VES value	String sent	Bytes of the String
1	“ 1”	0x20, 0x20, 0x20, 0x31
100	“ 100”	0x20, 0x31, 0x30, 0x30
>140	“>140”	0x3E, 0x31, 0x34, 0x30

H-FLAGS / L-FLAGS: Bitmap with 8-bit of the sample errors, represented in HEX-ASCII. The following table illustrates the errors:

Bit	Error	Description
0	Sample High	Blood column too high
1	Sample Low	Blood column too low
2	Sample Absent	Tube Empty
3	Abnormal	Error in acquisition of height
4	QC PASS	The VES of the QC measured is within the acceptability range
5	QC FAIL	The VES of the QC measured is outside the acceptability range
6-7	-	Reserved

EXAMPLES:

- In the case of a “Sample High” error the Bit 0 (least significant) will be set to one and all the others to zero, therefore the byte of the Flags will have a 0x01 hexadecimal value and its HEX-ASCII representation will be 0x30 / 0x31.
- In the case of a “QC Fail” error the Bit 5 will be set to one and all the others to zero, therefore the byte of the Flags will have a 0x20 hexadecimal value and its HEX-ASCII representation will be 0x32 / 0x30

RACK ID: string of 4 characters without terminator, identifies the classifier rack in which the sample has been repositioned.

POSITION: string of 2 characters without terminator, identifies the coordinates of the position in which the sample has been repositioned in the Classifier rack.

ATTENTION: for debugging purposes it is possible to disable the checksum control, replacing the H-COM bytes with the value of 0x44 instead of 0x35. In this case the two

bytes of the checksum are still sent but their value will be insignificant. The Host computer must also manage any possible cases in which the checksum is disabled.

